S0 Provisional per-model hypothesis decisions

Wave-pinning

Primary non-forced: H1 inconclusive; H2 inconclusive. Evaluable 8; gain CIs exclude 1: 3; reversal: none observed; f½ small CI [0.0001, 0.03], large CI [0.0001, 0.1]; f½ increase CIs separate: no.
Forced-inclusive: H1 inconclusive; H2 inconclusive. Evaluable 9; gain CIs exclude 1: 4; reversal: none observed; f½ small CI [0.0001, 0.03], large CI [0.0001, 0.1]; f½ increase CIs separate: no.

Goryachev

Primary non-forced: H1 inconclusive; H2 inconclusive. Evaluable 4; gain CIs exclude 1: 0; reversal: none observed; f½ small CI [0.0001, ∞], large CI [0.0001, ∞]; f½ increase CIs separate: no.
Forced-inclusive: H1 inconclusive; H2 inconclusive. Evaluable 5; gain CIs exclude 1: 1; reversal: none observed; f½ small CI [0.0001, 0.01], large CI [0.0001, 0.03]; f½ increase CIs separate: no.

Holmes 3

Primary non-forced: H1 inconclusive; H2 model criterion provisionally met. Evaluable 8; gain CIs exclude 1: 2; reversal: none observed; f½ small CI [0.0001, 0.03], large CI [0.0001, 0.03]; f½ increase CIs separate: no.
Forced-inclusive: H1 inconclusive; H2 model criterion provisionally met. Evaluable 9; gain CIs exclude 1: 3; reversal: none observed; f½ small CI [0.0001, 0.03], large CI [0.0001, 0.03]; f½ increase CIs separate: no.

Spring

Primary non-forced: H1 inconclusive; H2 inconclusive. Evaluable 7; gain CIs exclude 1: 4; reversal: none observed; f½ small CI [0.0001, 0.003], large CI [0.0001, 0.01]; f½ increase CIs separate: no.
Forced-inclusive: H1 criterion not met; H2 model criterion provisionally met. Evaluable 9; gain CIs exclude 1: 4; reversal: 1; f½ small CI [0.0001, 0.003], large CI [0.0001, 0.01]; f½ increase CIs separate: no.

S0 Decision memo and possible later framings

Provisional H1

Inconclusive in all four models; weakest-input persistence remains censored, fhalf intervals overlap, and empirical persistence ratios at weak input are not identified.

Provisional H2

Holmes satisfies the model peak-drop criterion in the main view, but it becomes inconclusive in the second-half-of-analysis and late-quarter windows. No non-forced evaluable condition has a ratio CI wholly below one. Overall H2 remains pending A6 and evidence in a second model. Holmes' criterion is not robust across all declared windows: second half, late quarter remain inconclusive.

Physical scope

Simulated polarity heading and qualified site mobility, not observed migration or an assumed cell-velocity law.

Model-dependent attenuation of the size advantage

Support: Holmes ratio falls from 2.77 at f=.03 to1.24 at .3 with large-cell multiplicity above .5; Spring ratios also decrease across .01-.1.
Weakness: No overall H2 decision; weak-input ratios and fhalf remain censored. Needed: E2 and A6; retain full regime maps.

Enlargement increases persistence over a measured intermediate-input range

Support: Non-forced ratio CIs exceed one in WP, Holmes and Spring at several strengths. Weakness: H1 is not established at every evaluable f and fhalf cannot be resolved from weak baselines. Needed: E4 plus adequate weak-input replication.

Multiplicity versus contrast loss under stronger input

Support: Large WP/Holmes/Spring become multiple-front dominated; large Gory becomes low contrast. Weakness: Gory strong-input means are forced, so H3 requires E1 and A5 fingerprints. Needed: Matched-mean E1 sigma sweep and optical A5/A6.

S1 Design, forcing and complete existing-case inventory

$$I = fI_{\mathrm{ref}}\exp(\sigma Z - \sigma^2/2), \quad \sigma = 1 \tag{1}$$

Four native ring models; chemical C = 20, 40, 60; Spring resting C = 1, 2, 3.

Original Gaussian-OU spatial forcing retained. Kernel length 1.5 (Spring 0.15); relaxation time 50 (Spring 5).

Native horizons 32,000 (Spring 3,200); discard the first half. Heading/scalar cadence 5 (Spring 0.5).

Saved profiles are coarser: 40 (Spring 4). E0 region lifetimes and peak-position sensitivities have that resolution.

Cosine endpoints require R ≥ 0.1. MSD tracks never bridge undefined headings. CIs resample whole independent seeds (2,000 draws).

Persistence is the empirical first 1/e crossing. Upward triangles denote finite-window lower bounds; downward triangles denote cadence limits.

Grey indicates an abnormal matched-mean control. Missing forced-state classification stays unavailable.

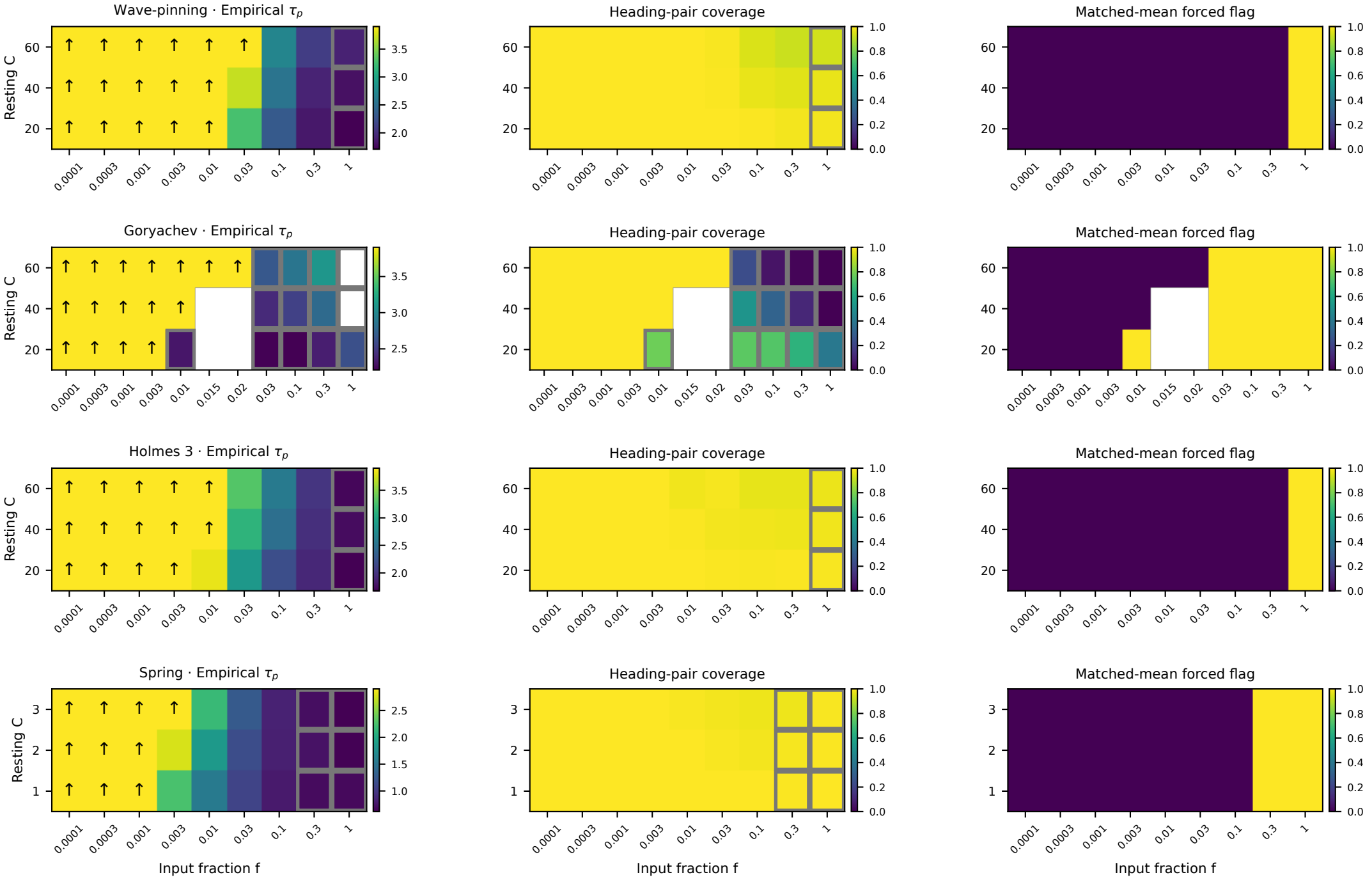
Spring physical increments use the measured circumference at each frame. Native times are compared only within a model.

Spring's forcing stays attached to material nodes: its physical covariance width is 0.15 × C(t)/C(basal), rather than a retuned fixed optical width.

Blue: smallest ring; green: intermediate ring; orange: largest ring. Grey: forced matched-mean condition, retained in every display.

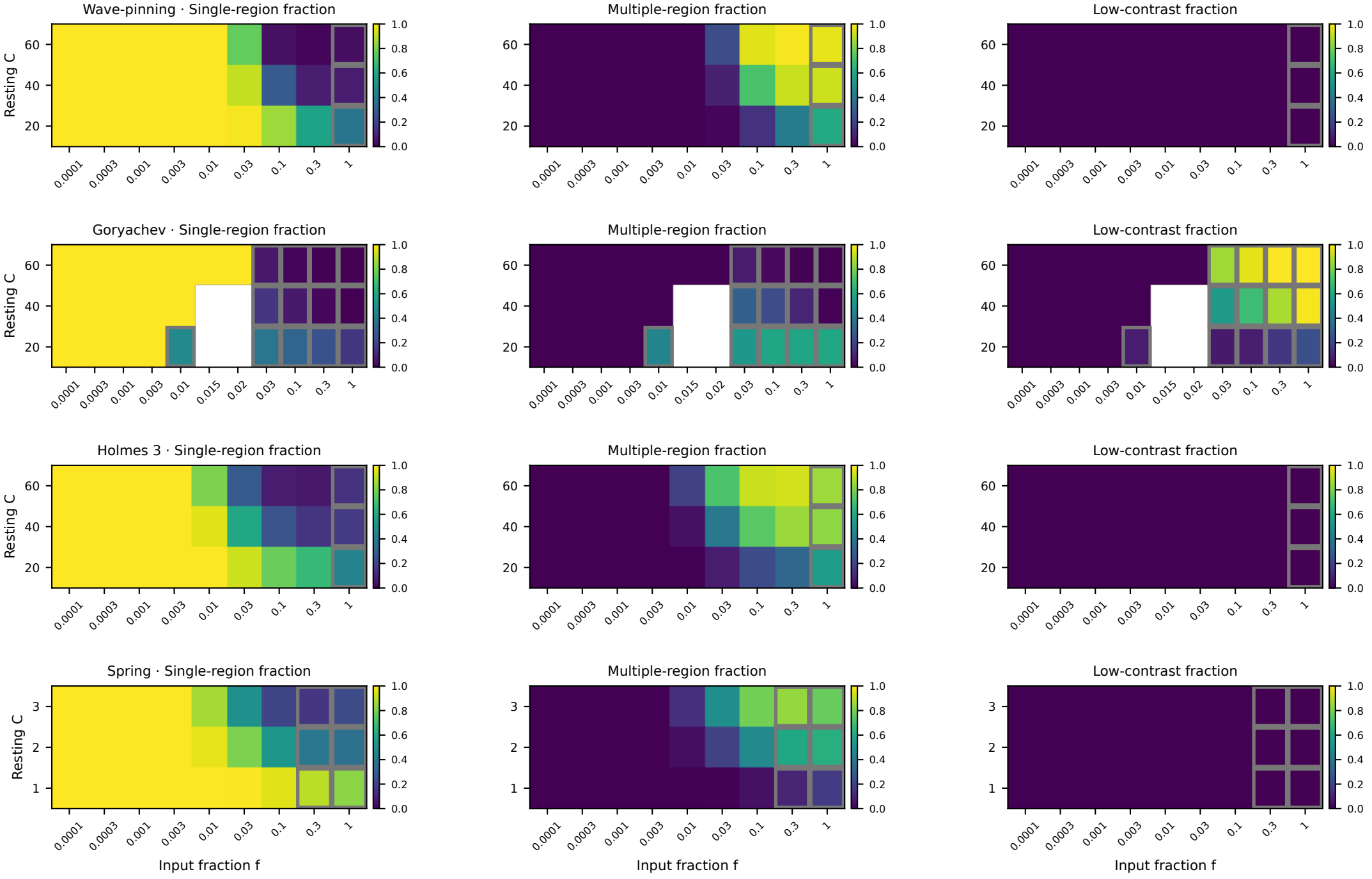
Model	Noisy / mean / zero	Seeds / condition	Observed input fractions f
Wave-pinning	204/27/3	2, 12	0.0001, 0.0003, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1
Goryachev	228/29/3	2, 12	0.0001, 0.0003, 0.001, 0.003, 0.01, 0.015, 0.02, 0.03, 0.1, 0.3, 1
Holmes 3	204/27/3	2, 12	0.0001, 0.0003, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1
Spring	204/27/3	2, 12	0.0001, 0.0003, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1

S2 Condition overview · log₁₀ native persistence



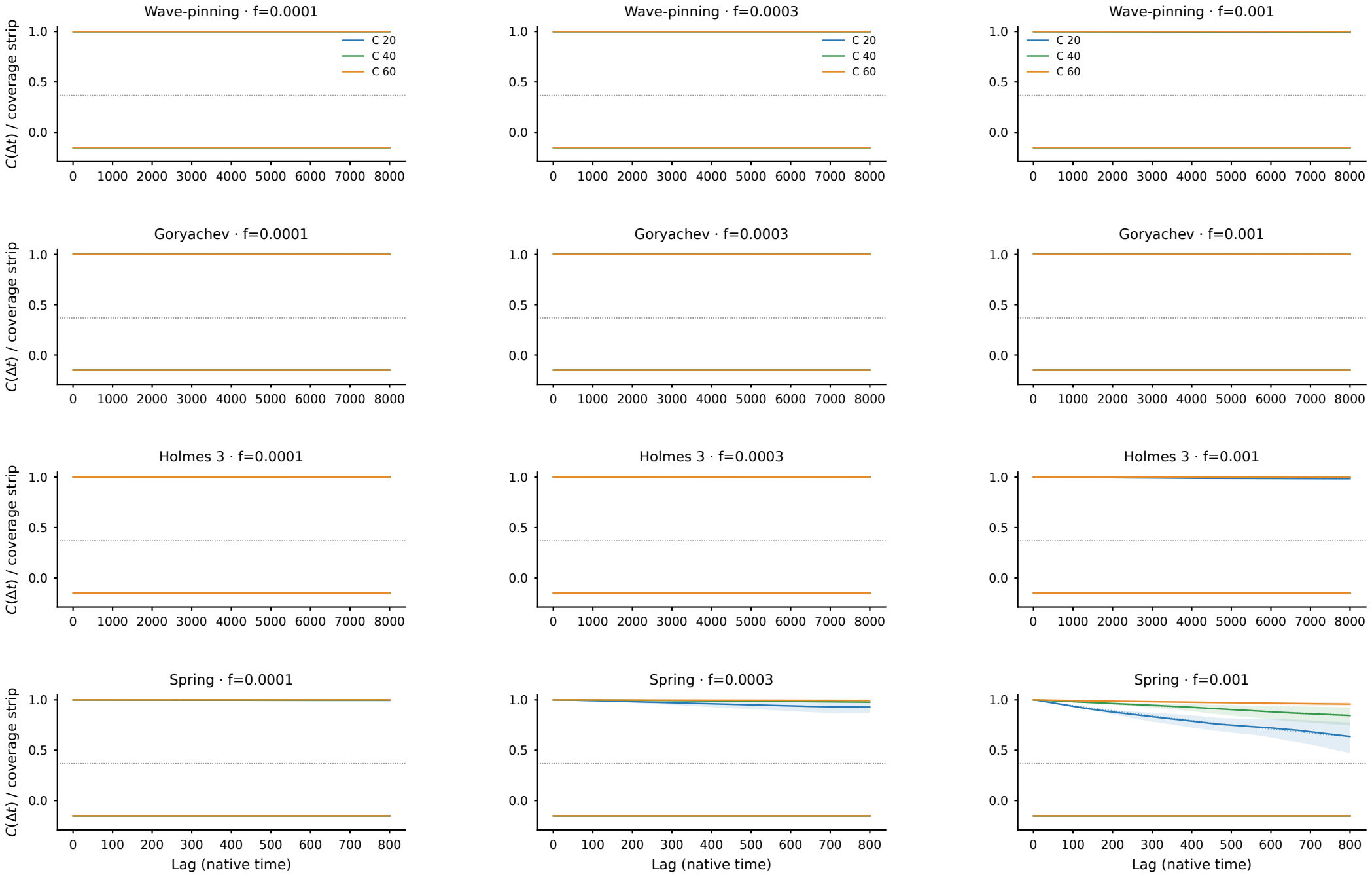
E0 INTERIM · A1–A4/A7 · provisional decisions · All observed strengths shown, including Goryachev-only refinement. Grey outlines flag forced conditions; arrows preserve censoring and cadence limits.
CSV: figure4_E0_04_S2_1.csv

S2 Condition overview · state fractions



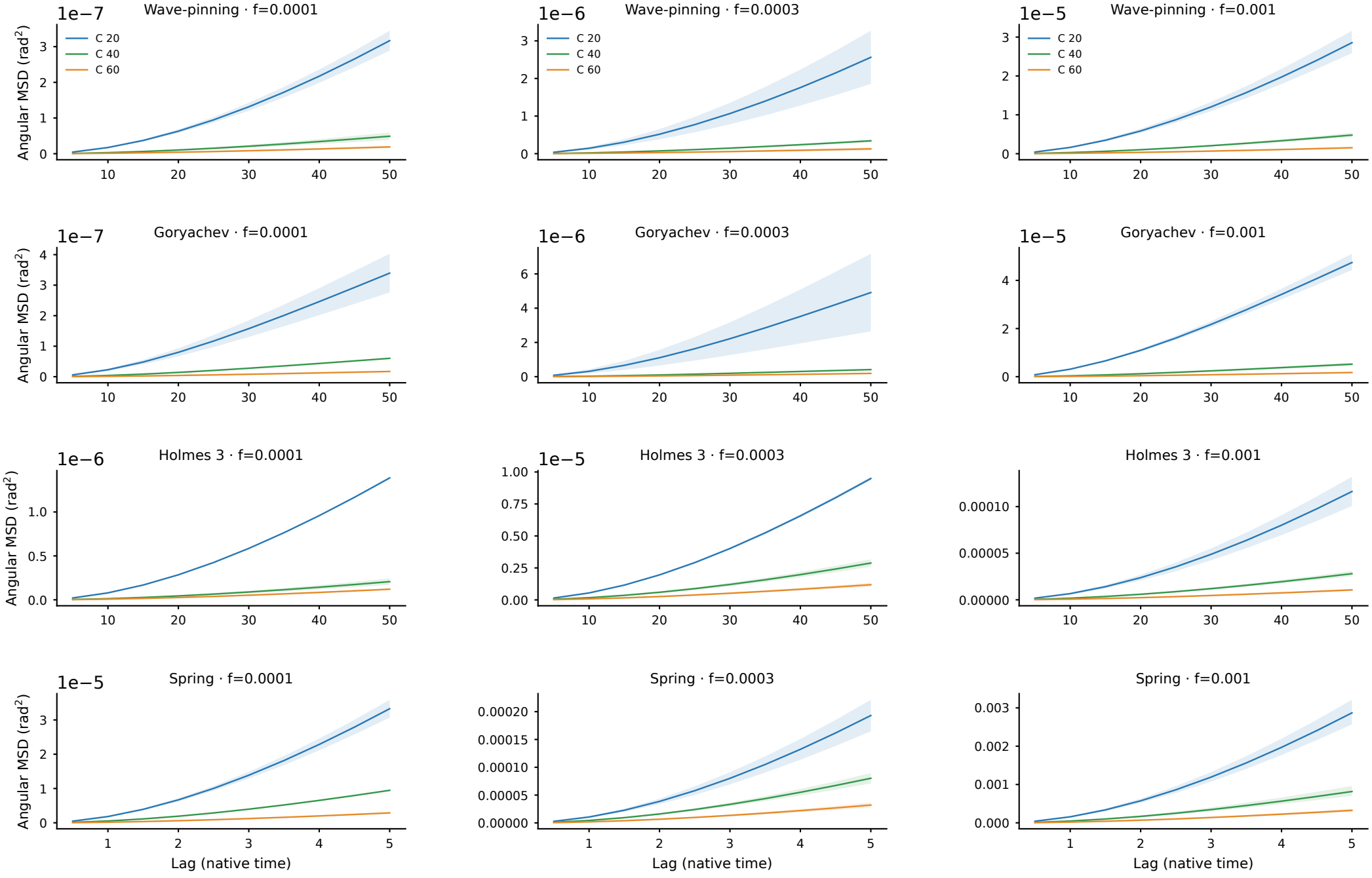
E0 INTERIM · A1–A4/A7 · provisional decisions · All observed strengths shown, including Goryachev-only refinement. Grey outlines flag forced conditions; arrows preserve censoring and cadence limits.
CSV: figure4_E0_05_S2_2.csv

S3 All heading correlations



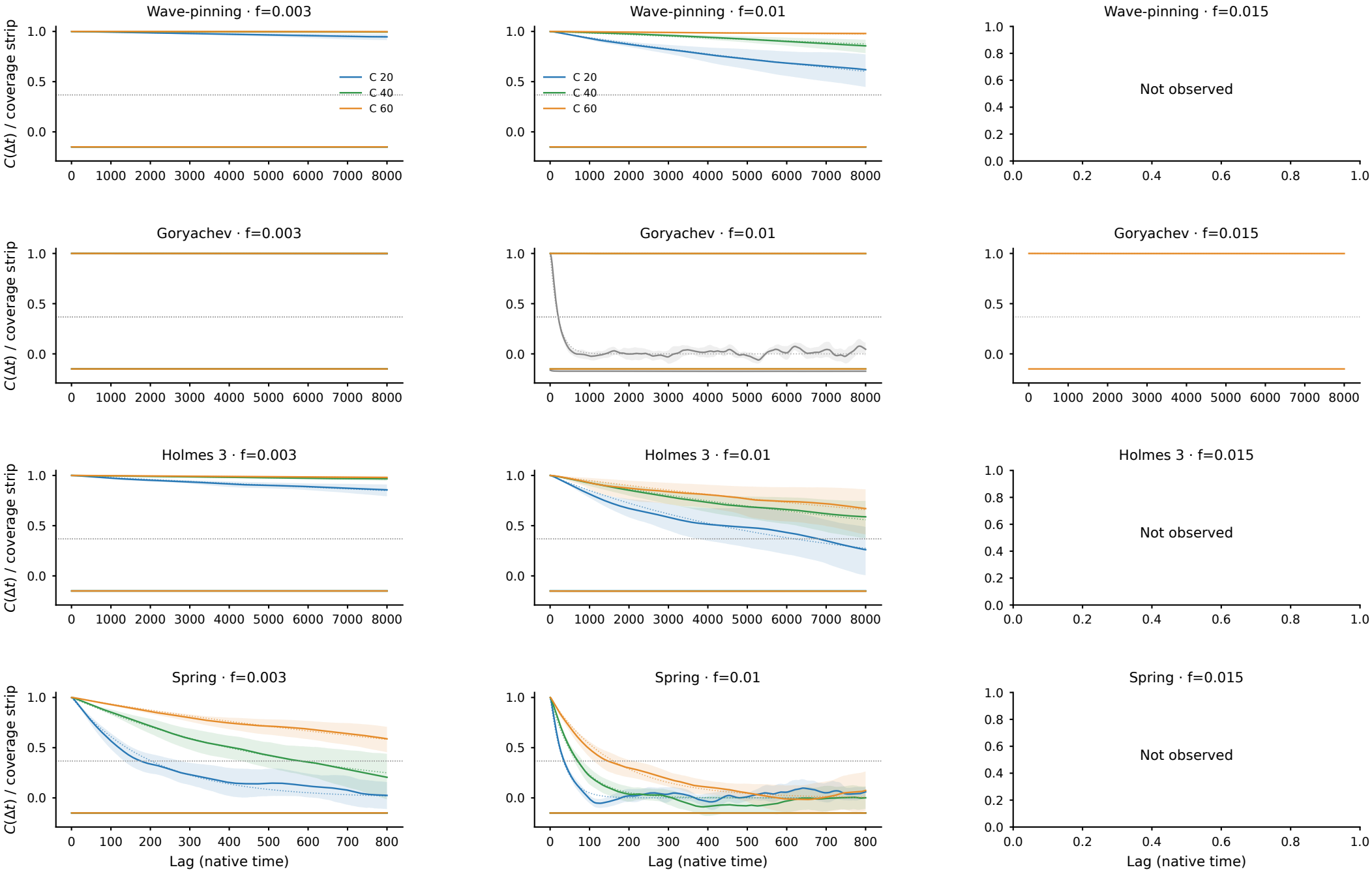
E0 INTERIM · A1–A4/A7 · provisional decisions · Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. Cosine coverage strips span −0.25 to −0.15 for coverage 0 to 1; MSD tracks never bridge invalid headings.
CSV: figure4_E0_06_S3_1_cosine.csv

S3 Short-lag angular MSD



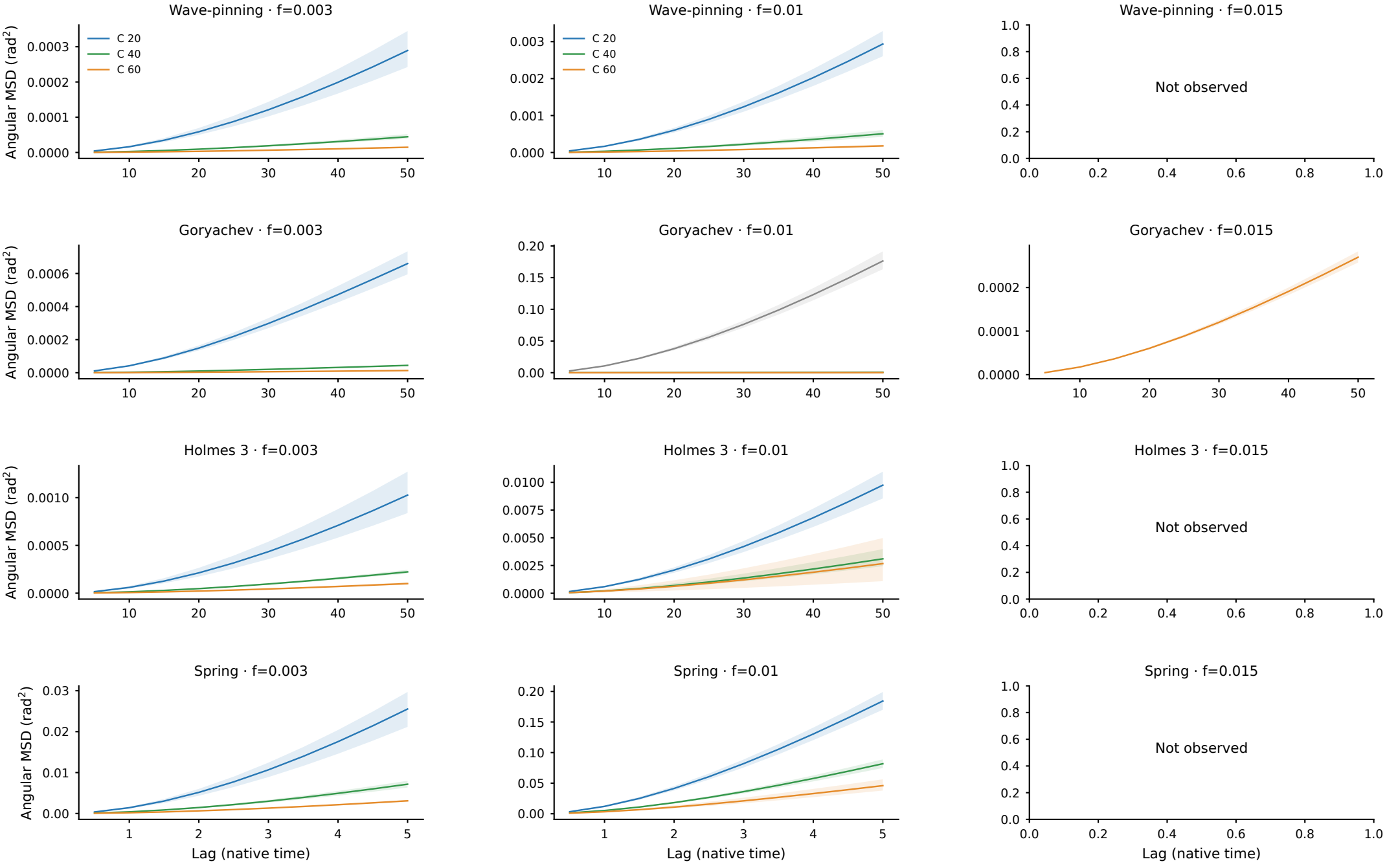
E0 INTERIM · A1–A4/A7 · provisional decisions · Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. Cosine coverage strips span −0.25 to −0.15 for coverage 0 to 1; MSD tracks never bridge invalid headings.
CSV: figure4_E0_07_S3_1_angular_msd.csv

S3 All heading correlations



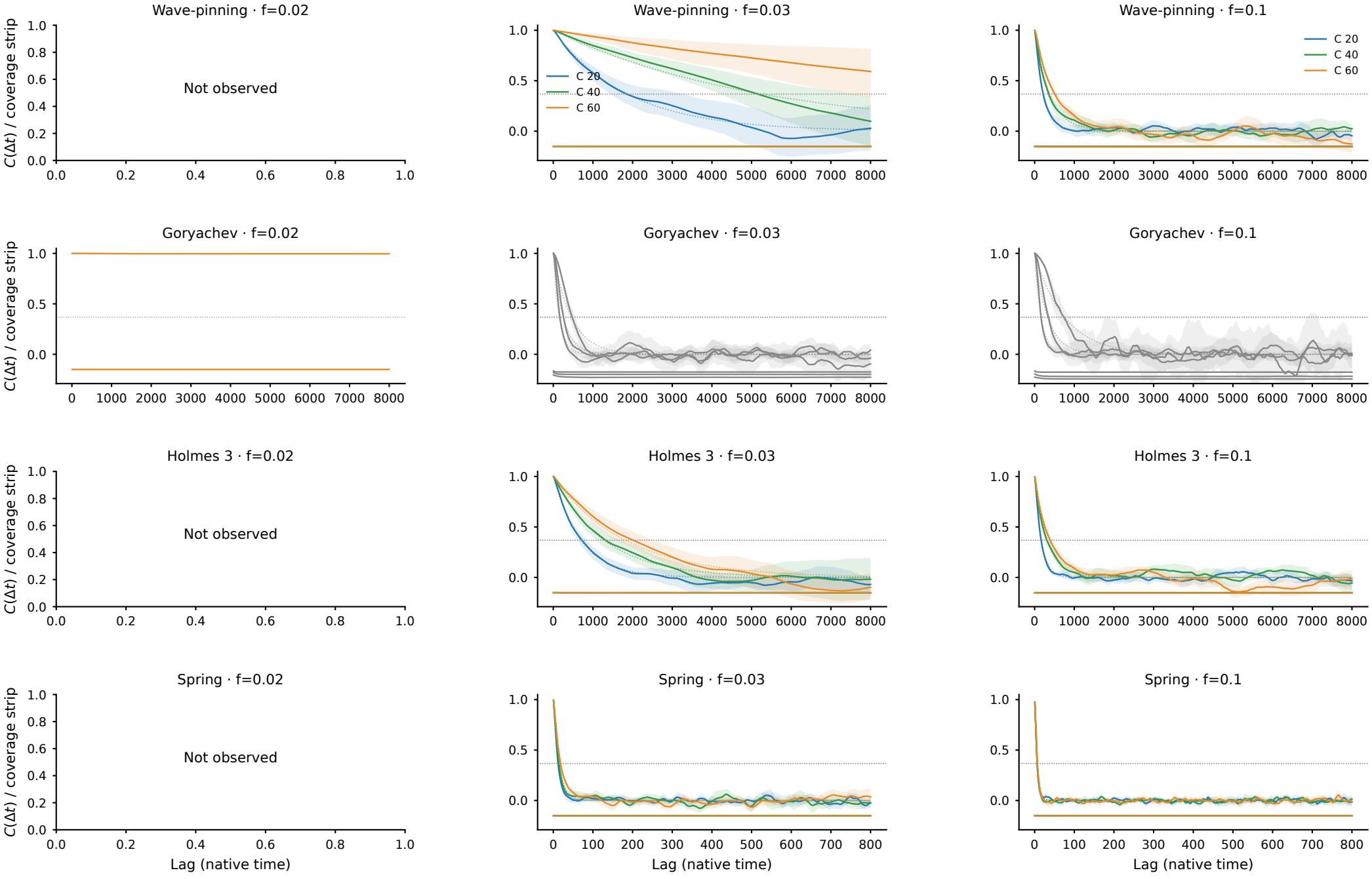
E0 INTERIM · A1–A4/A7 · provisional decisions · Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. Cosine coverage strips span −0.25 to −0.15 for coverage 0 to 1; MSD tracks never bridge invalid headings.
CSV: figure4_E0_08_S3_2_cosine.csv

S3 Short-lag angular MSD



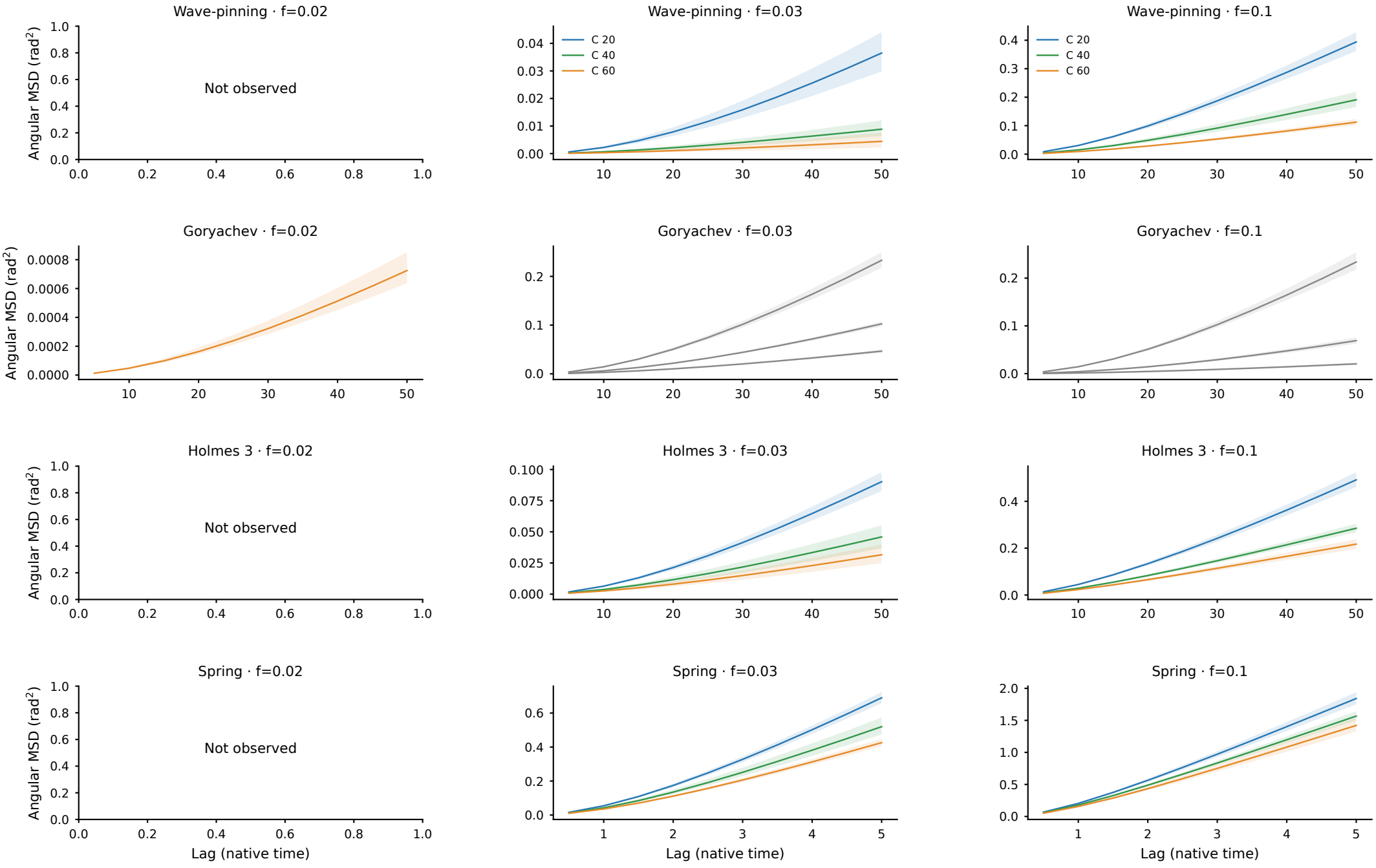
E0 INTERIM · A1–A4/A7 · provisional decisions · Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. Cosine coverage strips span −0.25 to −0.15 for coverage 0 to 1; MSD tracks never bridge invalid headings.
CSV: figure4_E0_09_S3_2_angular_msd.csv

S3 All heading correlations



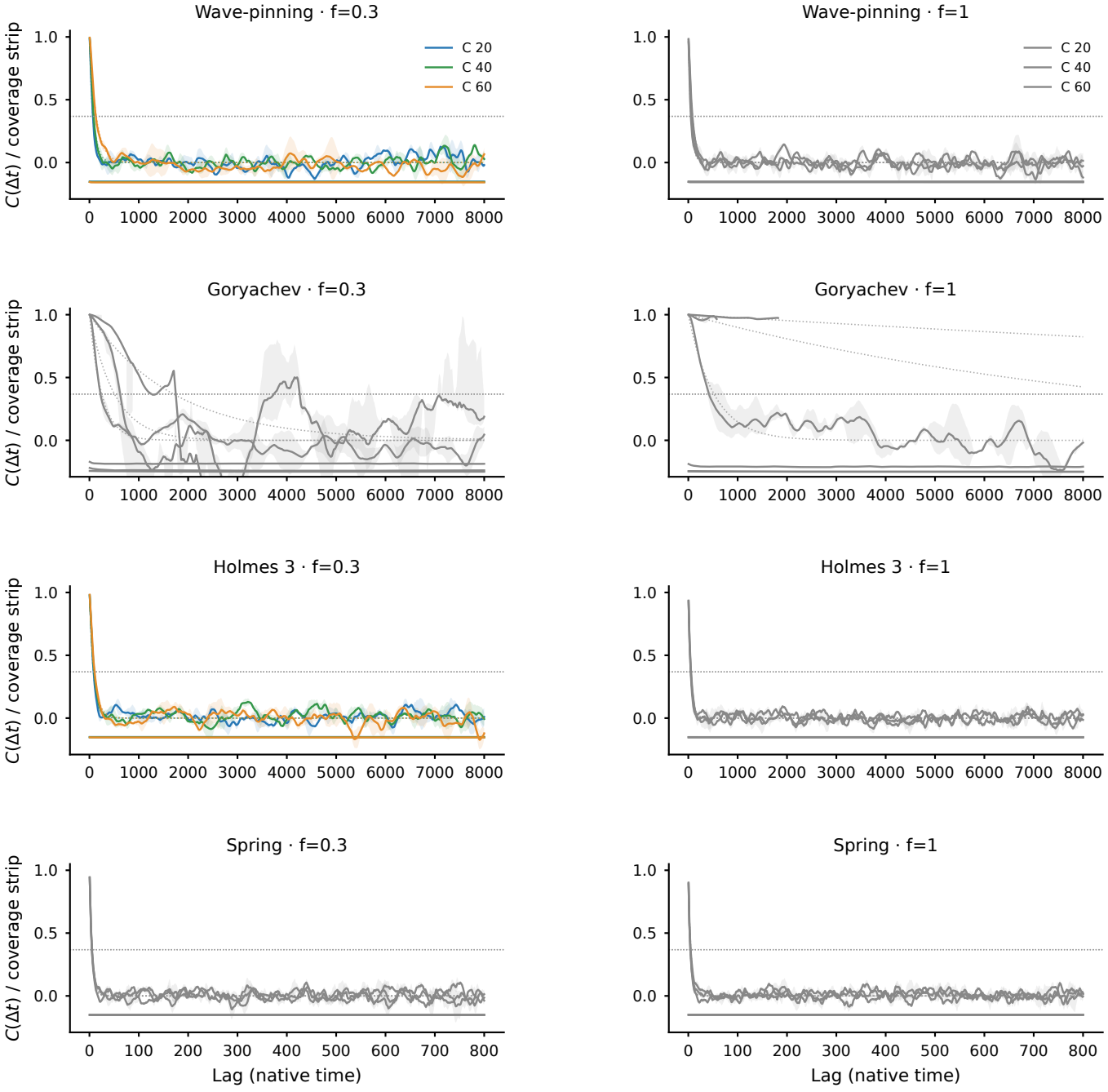
E0 INTERIM · A1–A4/A7 · provisional decisions · Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. Cosine coverage strips span −0.25 to −0.15 for coverage 0 to 1; MSD tracks never bridge invalid headings.
CSV: figure4_E0_10_S3_3_cosine.csv

S3 Short-lag angular MSD



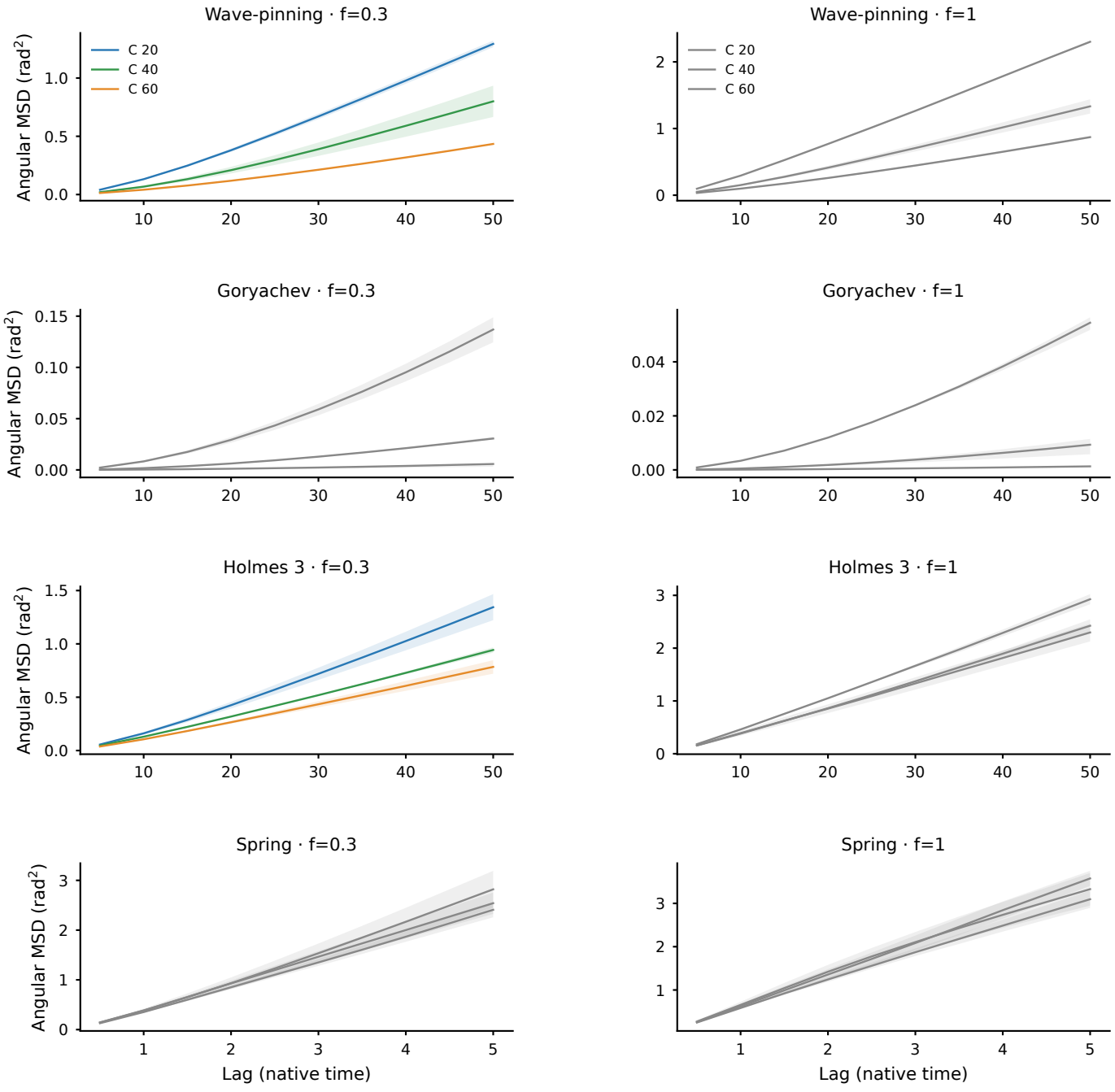
E0 INTERIM · A1–A4/A7 · provisional decisions · Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. Cosine coverage strips span −0.25 to −0.15 for coverage 0 to 1; MSD tracks never bridge invalid headings.
CSV: figure4_E0_11_S3_3_angular_msd.csv

S3 All heading correlations



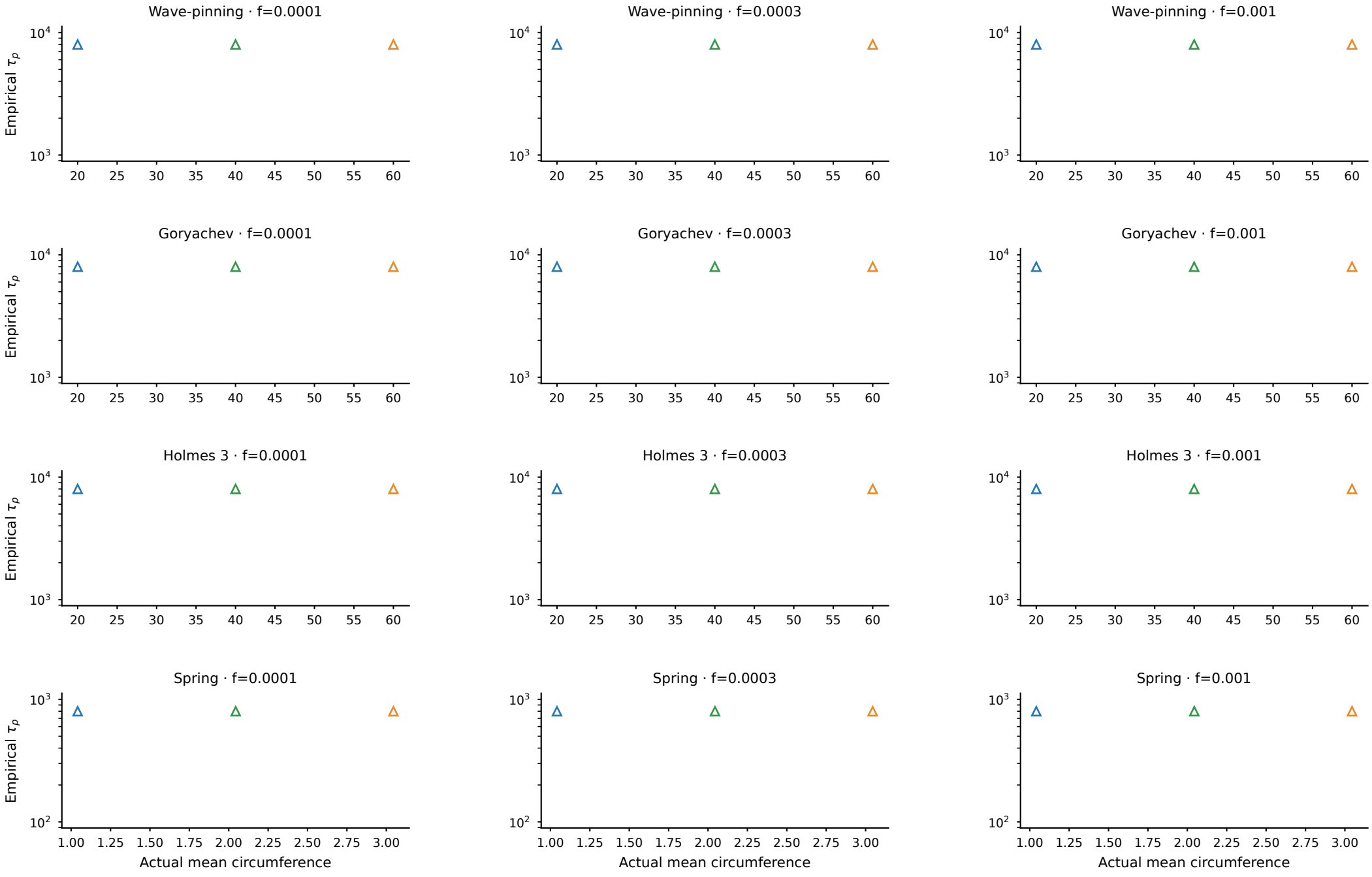
E0 INTERIM · A1–A4/A7 · provisional decisions · Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. Cosine coverage strips span −0.25 to −0.15 for coverage 0 to 1; MSD tracks never bridge invalid headings.
CSV: figure4_E0_12_S3_4_cosine.csv

S3 Short-lag angular MSD



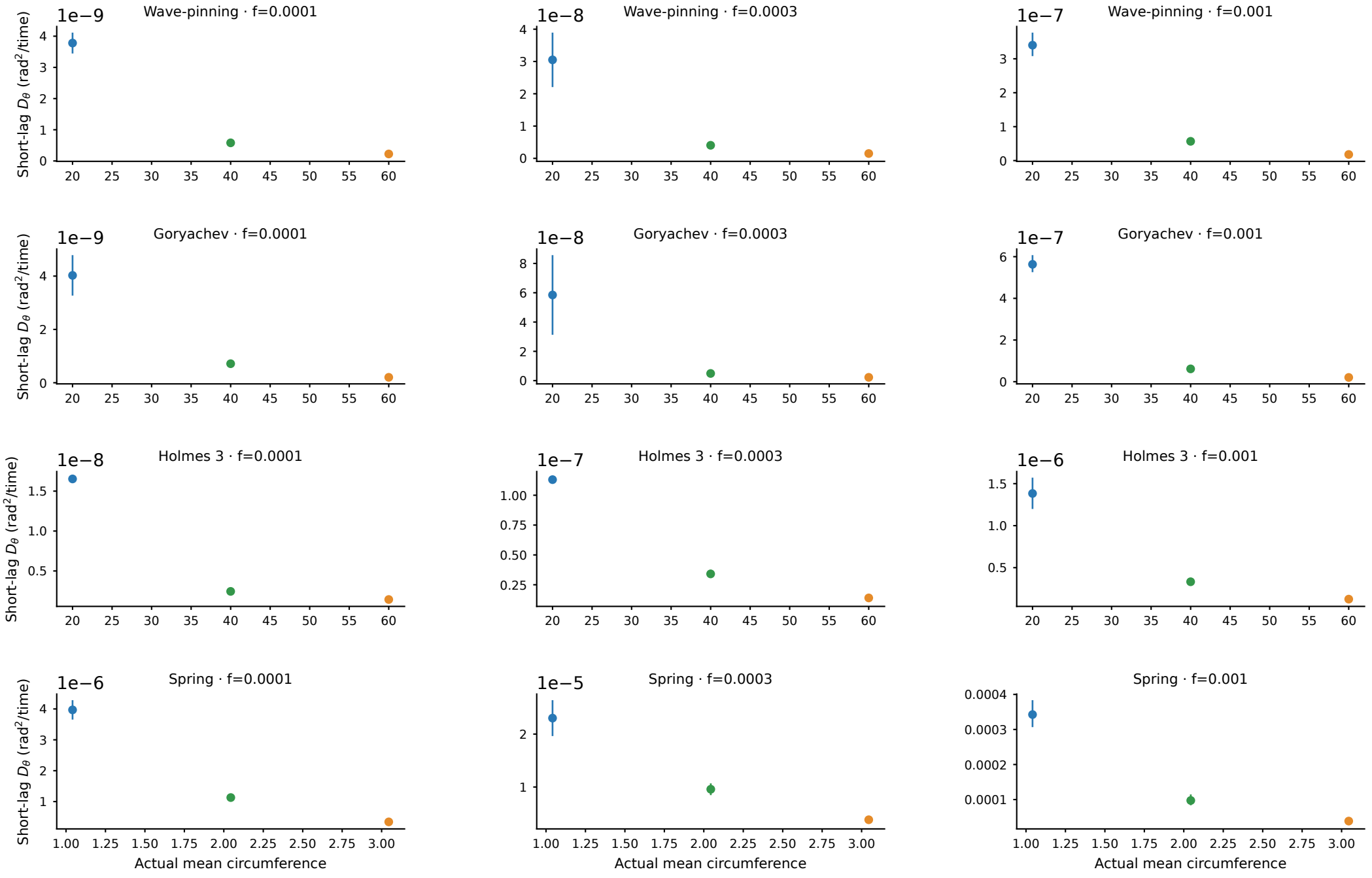
E0 INTERIM · A1–A4/A7 · provisional decisions · Lines/intervals aggregate whole seeds; dotted exponentials are comparisons, never empirical-bound replacements. All lag rows remain in CSV. Cosine coverage strips span −0.25 to −0.15 for coverage 0 to 1; MSD tracks never bridge invalid headings.
CSV: figure4_E0_13_S3_4_angular_msd.csv

S4 Persistence and angular mobility across size



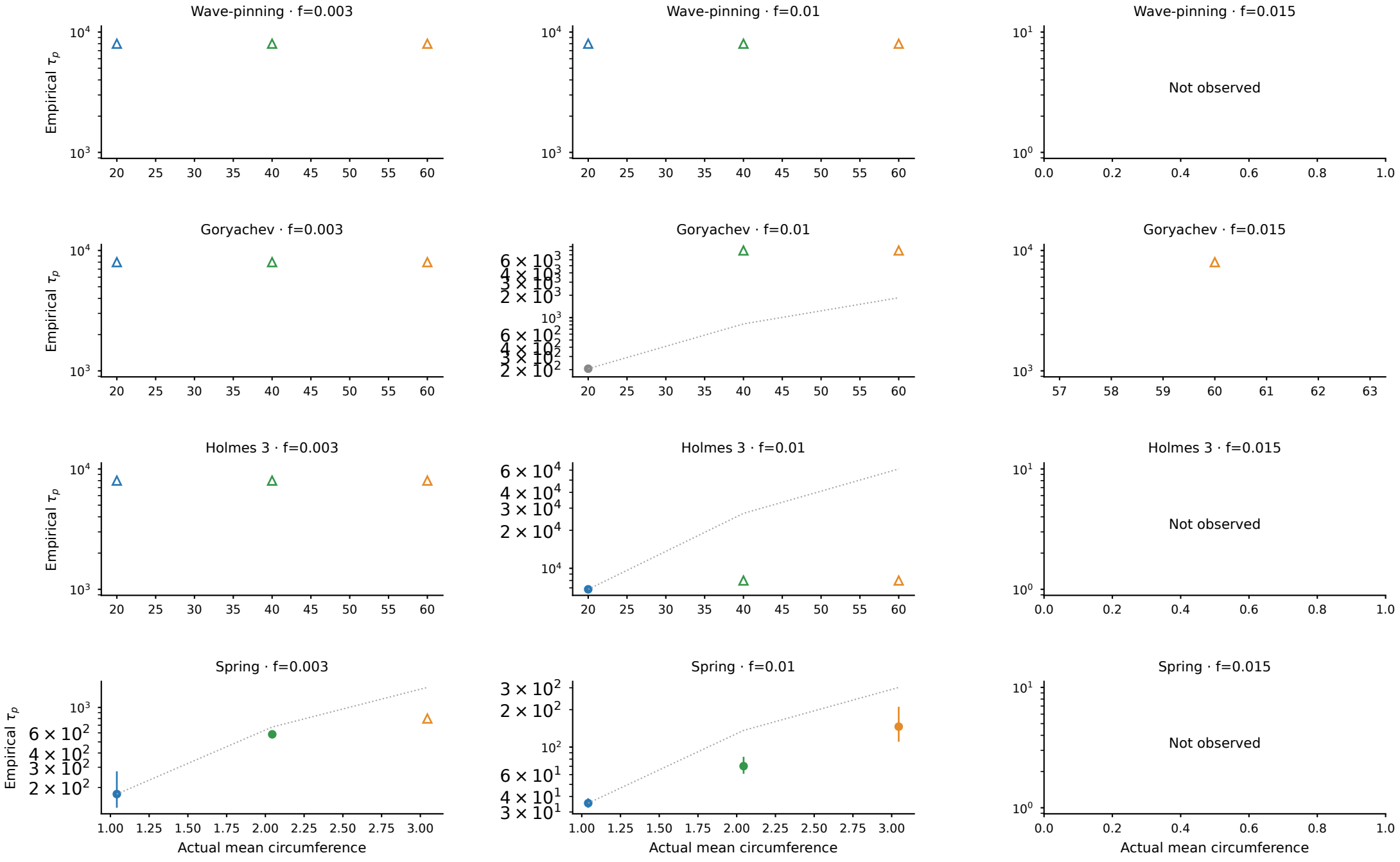
E0 INTERIM · A1–A4/A7 · provisional decisions · Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence.
CSV: figure4_E0_14_S4_1_tau_p.csv

S4 Persistence and angular mobility across size



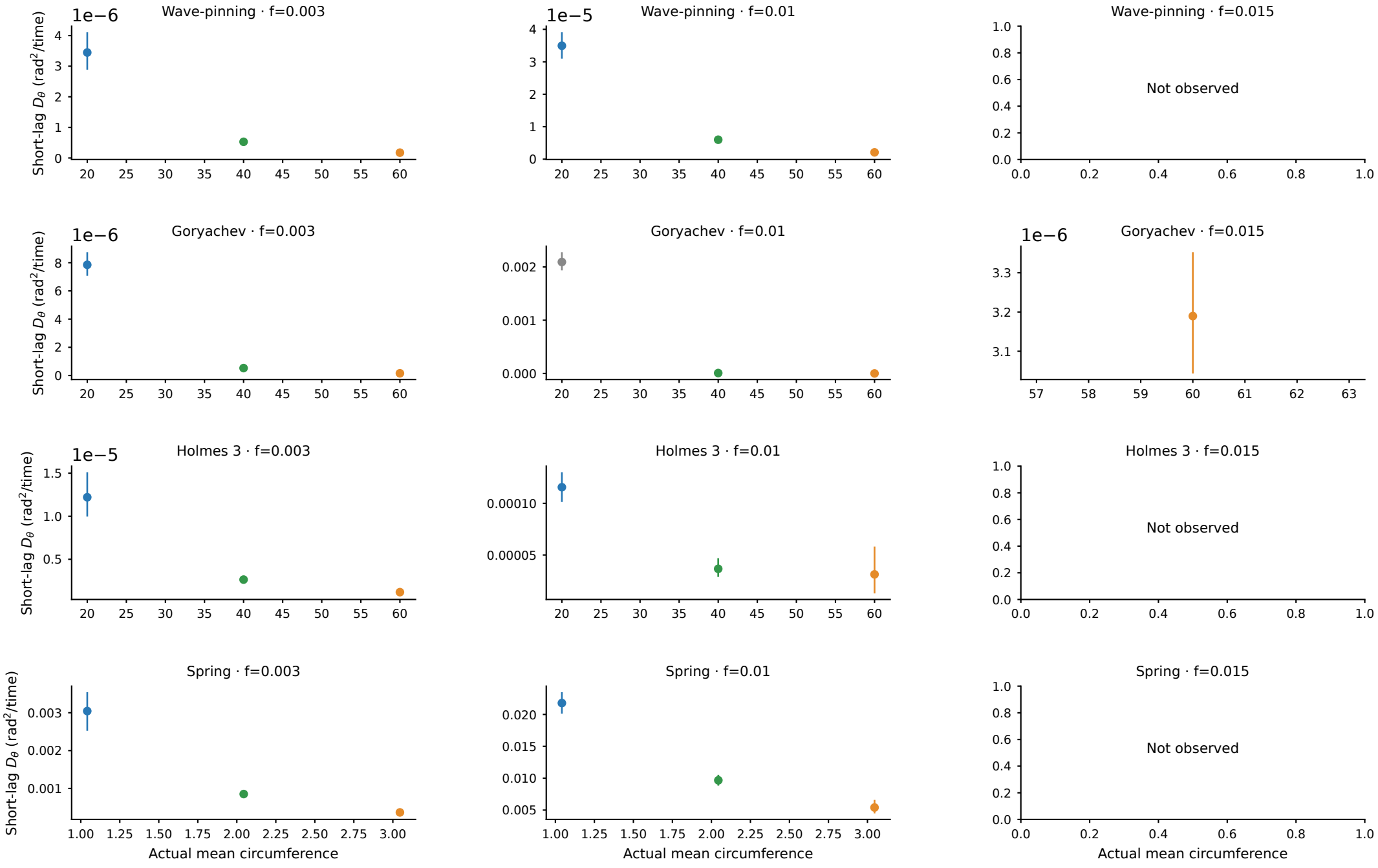
E0 INTERIM · A1–A4/A7 · provisional decisions · Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence.
CSV: figure4_E0_15_S4_1_Dtheta.csv

S4 Persistence and angular mobility across size



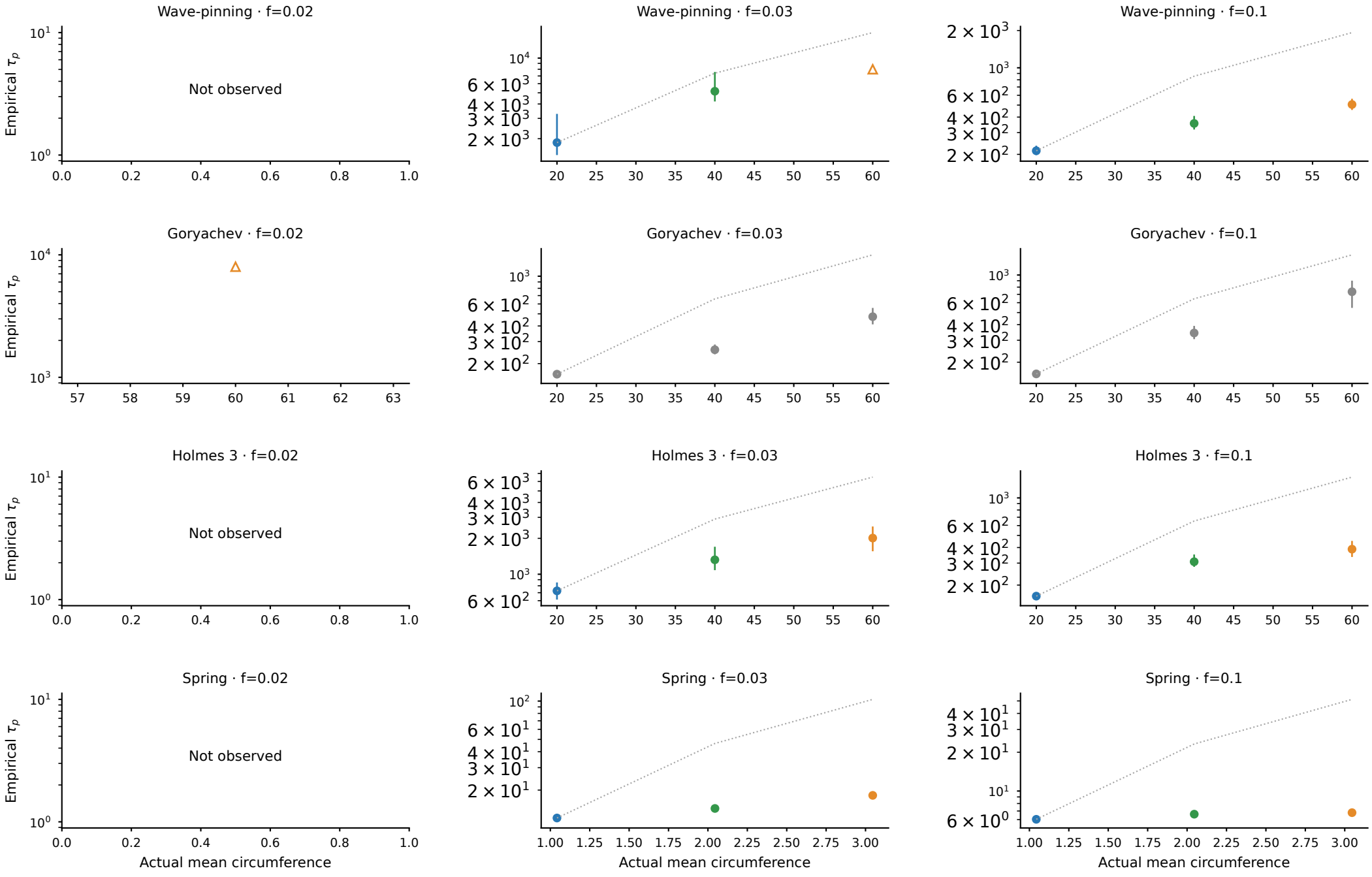
E0 INTERIM · A1–A4/A7 · provisional decisions · Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence.
CSV: figure4_E0_16_S4_2_tau_p.csv

S4 Persistence and angular mobility across size



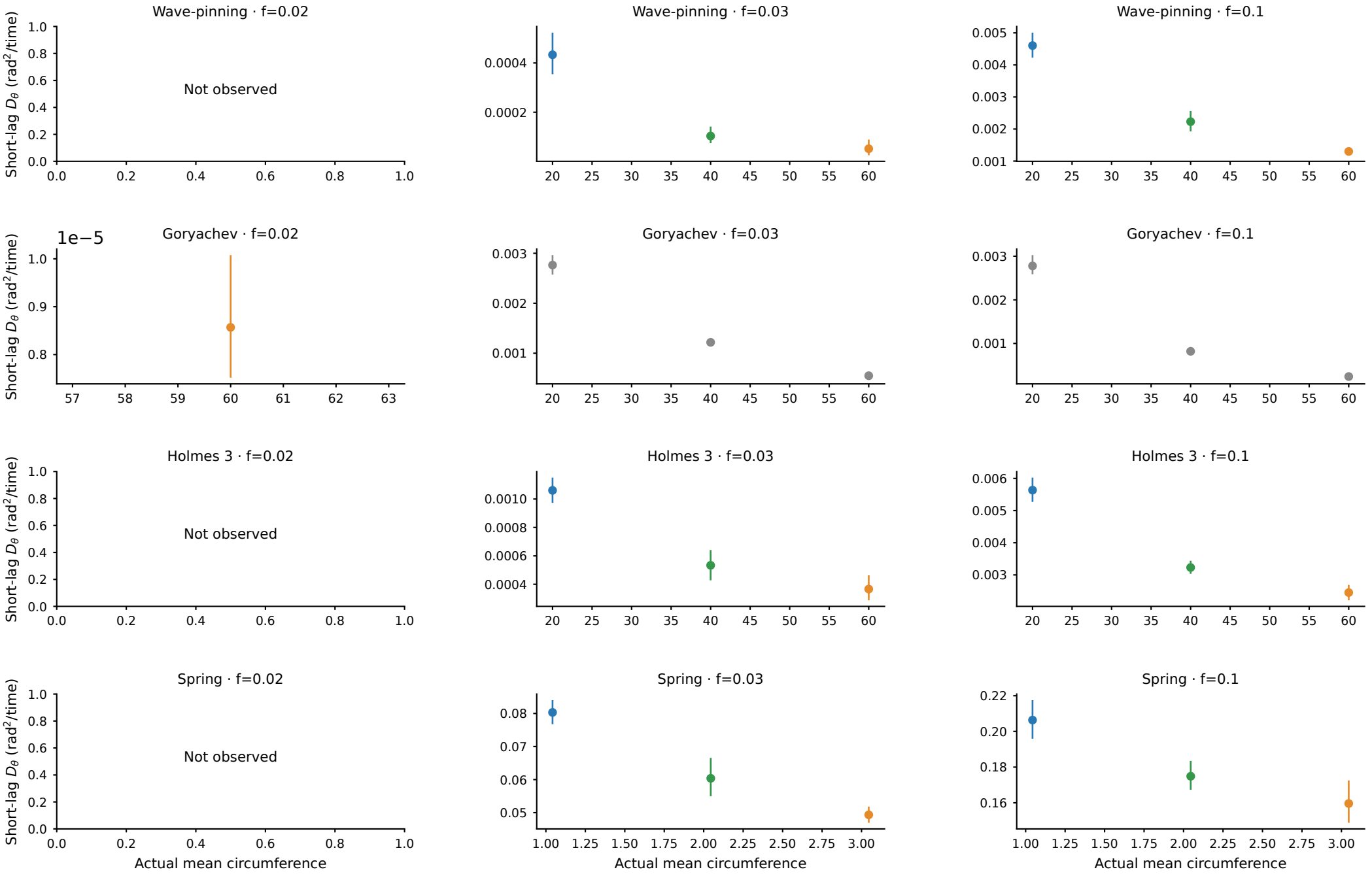
E0 INTERIM · A1–A4/A7 · provisional decisions · Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence.
CSV: figure4_E0_17_S4_2_Dtheta.csv

S4 Persistence and angular mobility across size



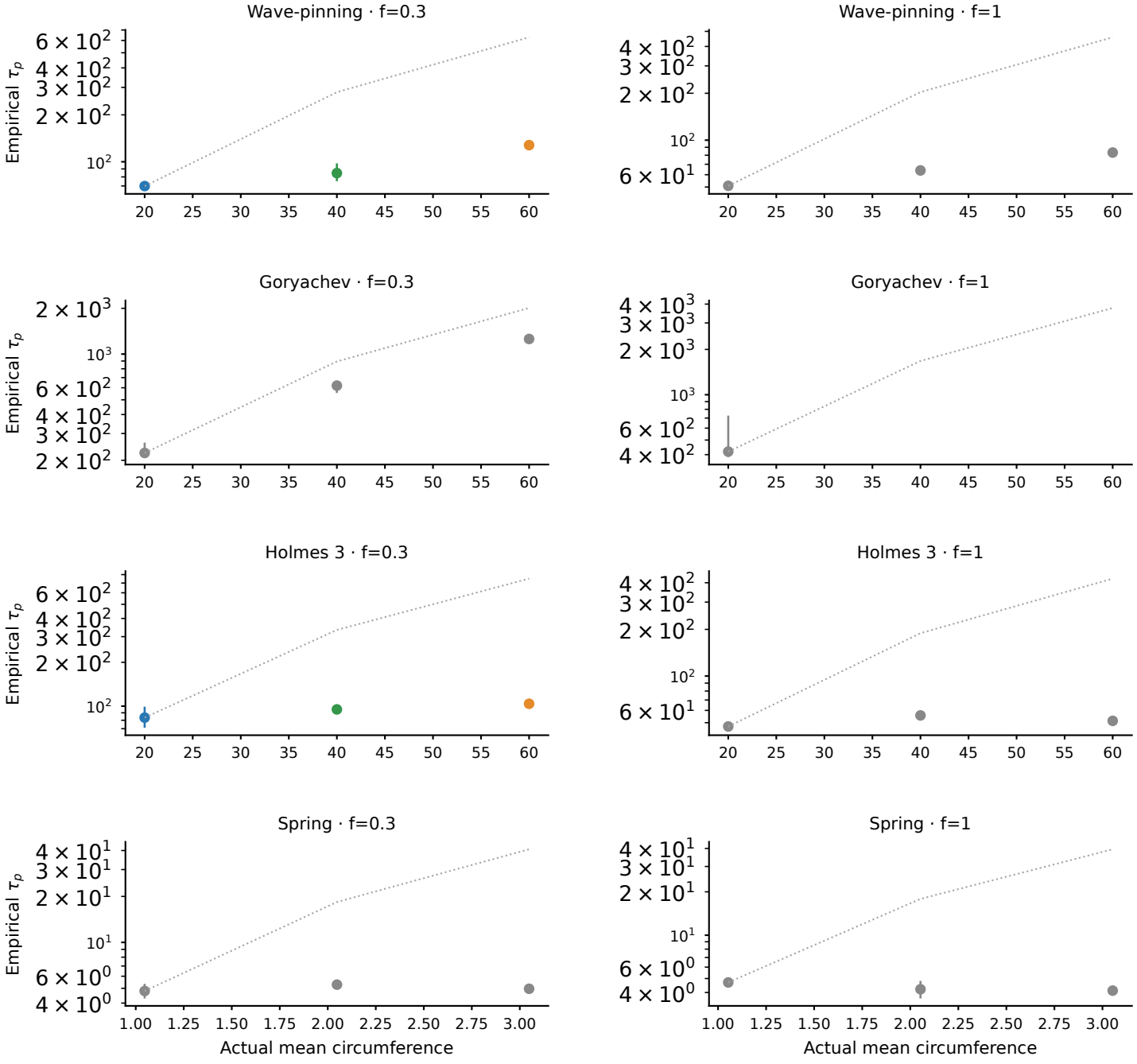
E0 INTERIM · A1–A4/A7 · provisional decisions · Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence.
CSV: figure4_E0_18_S4_3_tau_p.csv

S4 Persistence and angular mobility across size



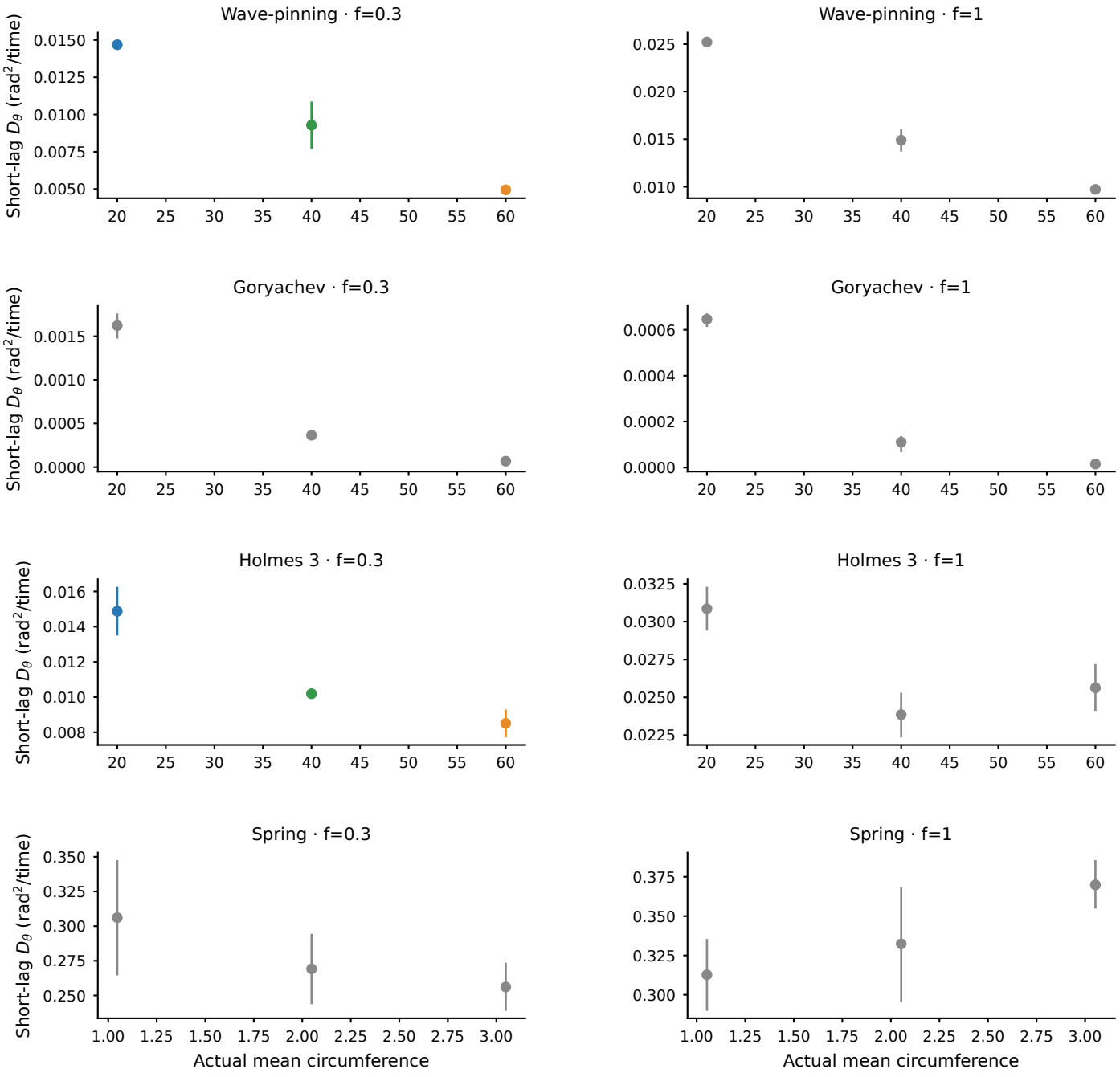
E0 INTERIM · A1–A4/A7 · provisional decisions · Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence.
CSV: figure4_E0_19_S4_3_Dtheta.csv

S4 Persistence and angular mobility across size



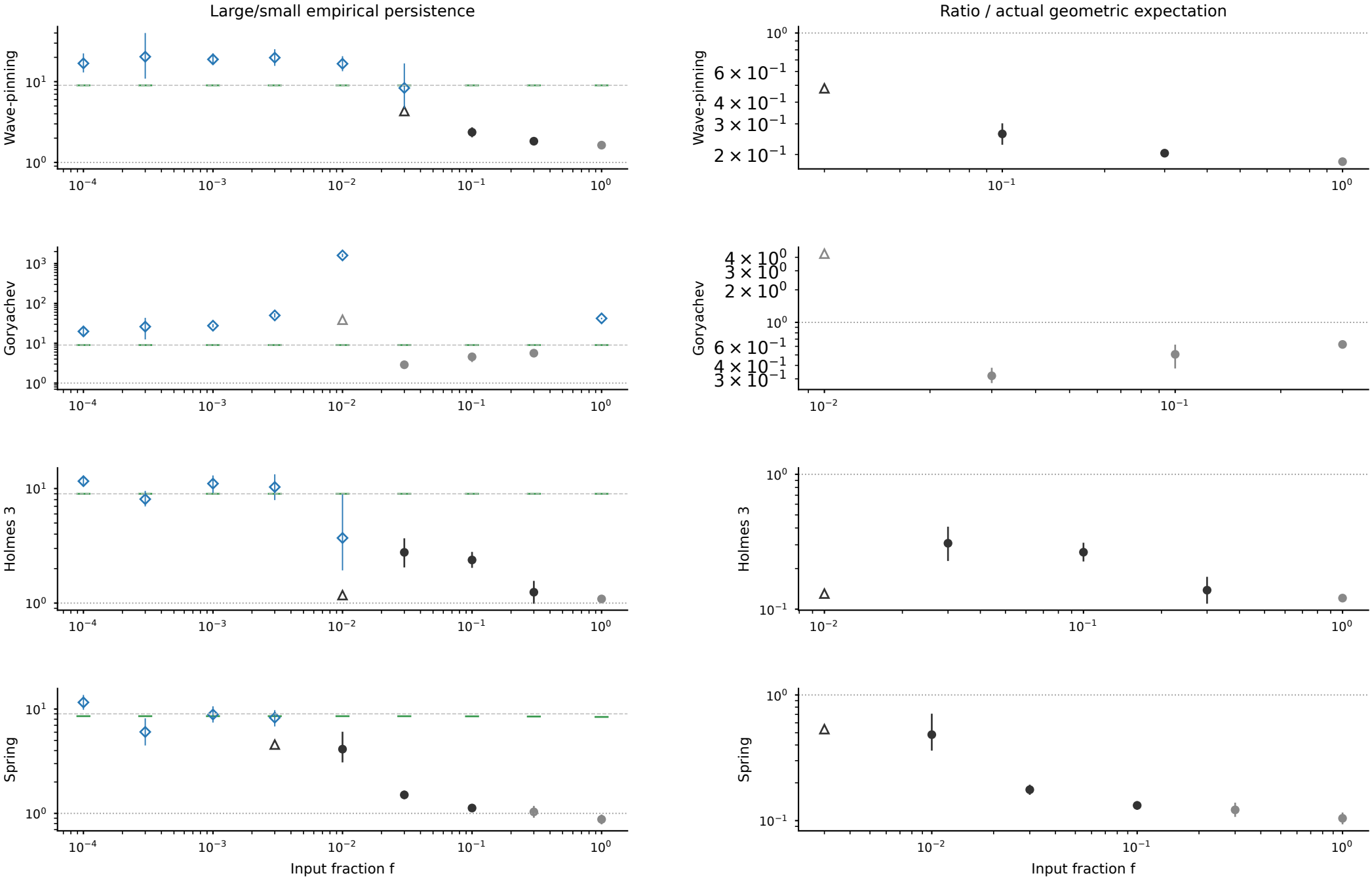
E0 INTERIM · A1–A4/A7 · provisional decisions · Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence.
CSV: figure4_E0_20_S4_4_tau_p.csv

S4 Persistence and angular mobility across size



E0 INTERIM · A1–A4/A7 · provisional decisions · Circumference is measured in the analysis window. Dθ is a signed short-lag mobility estimate, not an assumption of free diffusion. Dotted C² reference is anchored only to a resolved smallest-size persistence.
CSV: figure4_E0_21_S4_4_Dtheta.csv

S4 Size advantage and geometric reference



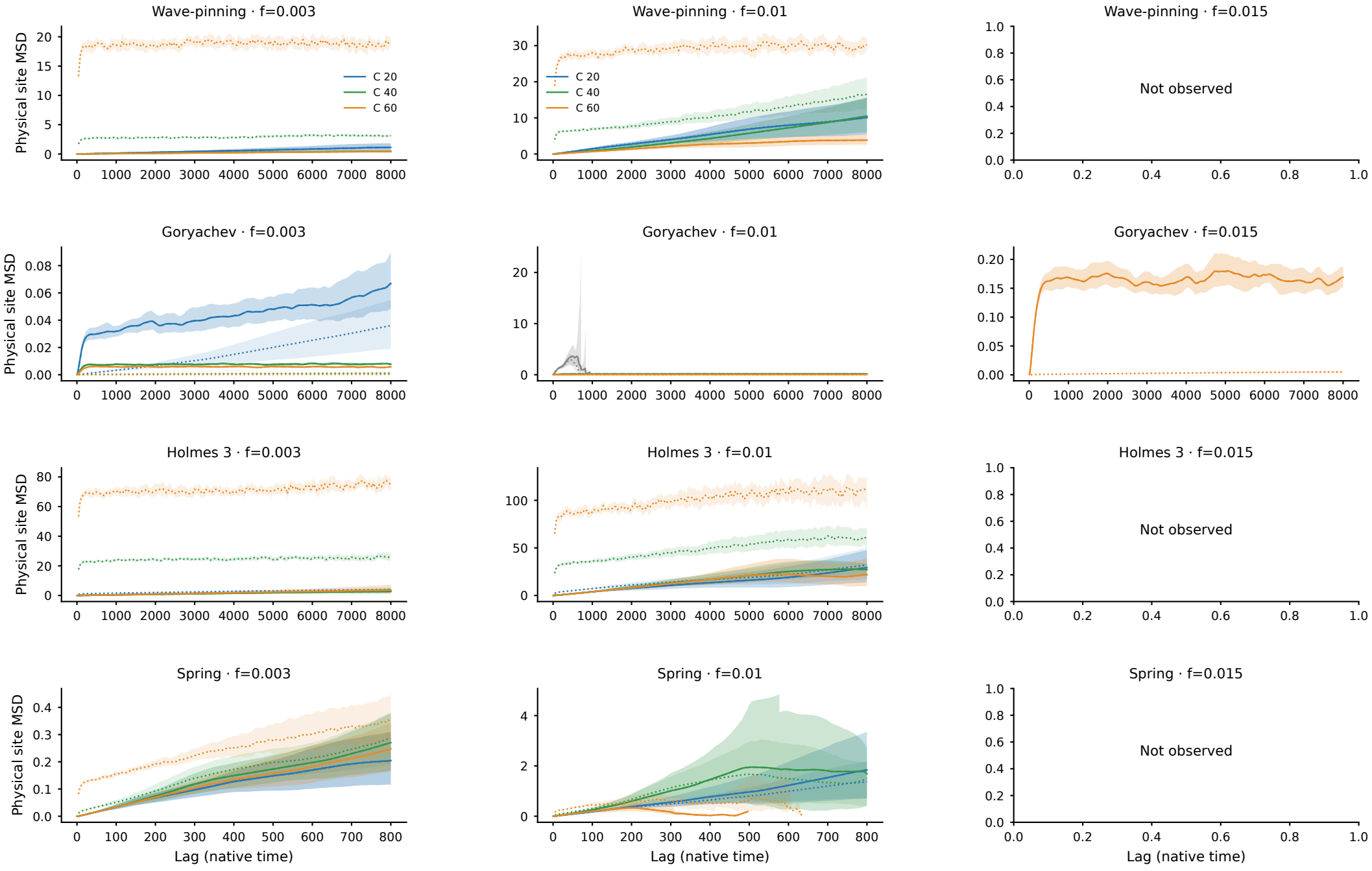
E0 INTERIM · A1–A4/A7 · provisional decisions · Triangles preserve finite-window or resolution bounds. Two censored times do not yield a finite ratio.
Open blue diamonds: small/large Dθ mobility proxy when the time ratio is unresolved. Green marks: measured circumference-ratio squared; dashed 9: resting-size reference.
CSV: figure4_E0_22_S4_ratios.csv

S5 Physical site motion in single-qualified-region tracks



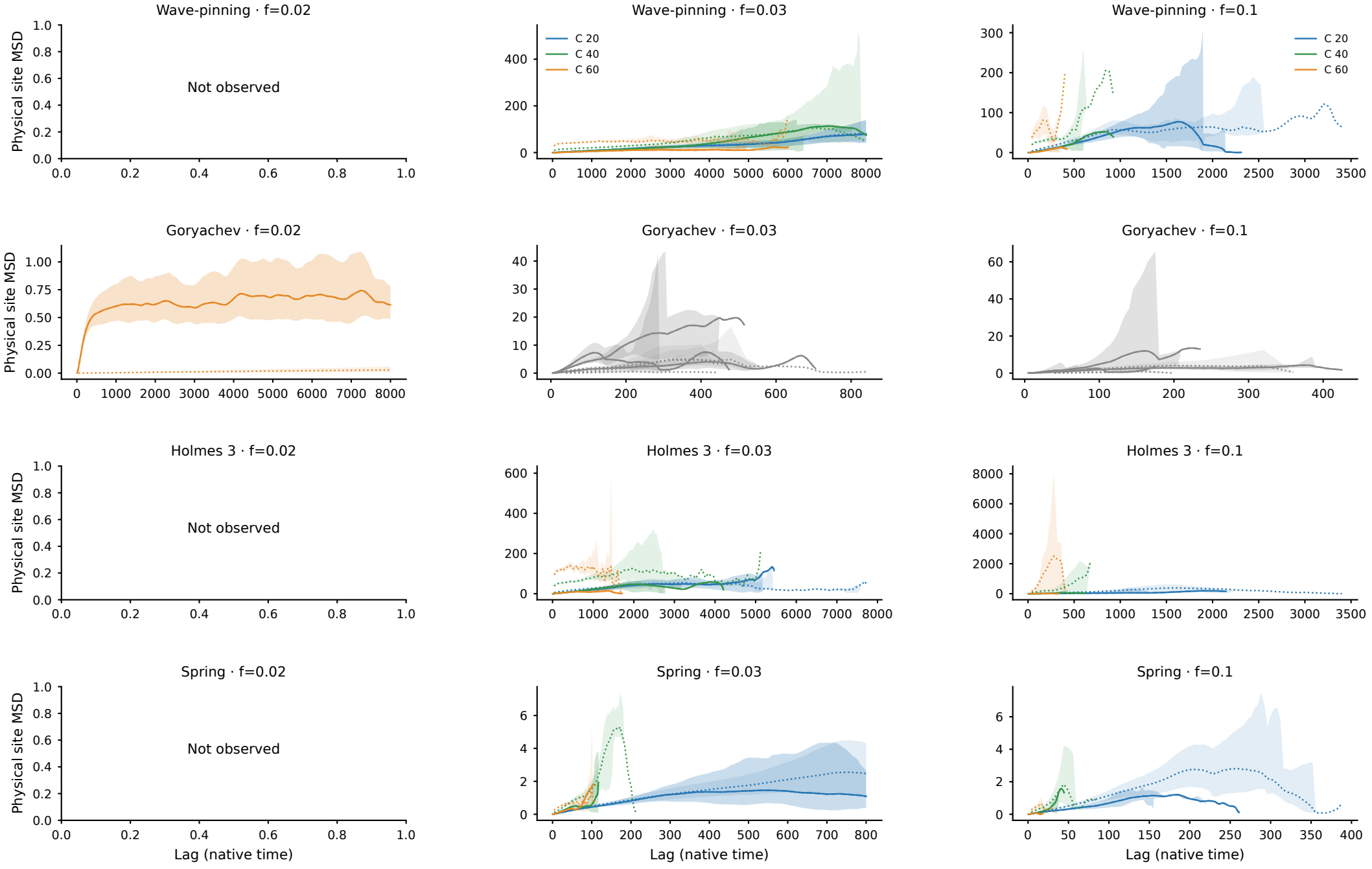
E0 INTERIM · A1–A4/A7 · provisional decisions · Solid: midpoint-C arc increments. Dashed: direct angle×actual-C sensitivity. Dotted: profile-peak sensitivity at coarse cadence. All methods reset at invalid frames; a plateau can indicate confinement.
CSV: figure4_E0_23_S5_1_siteMSD.csv

S5 Physical site motion in single-qualified-region tracks



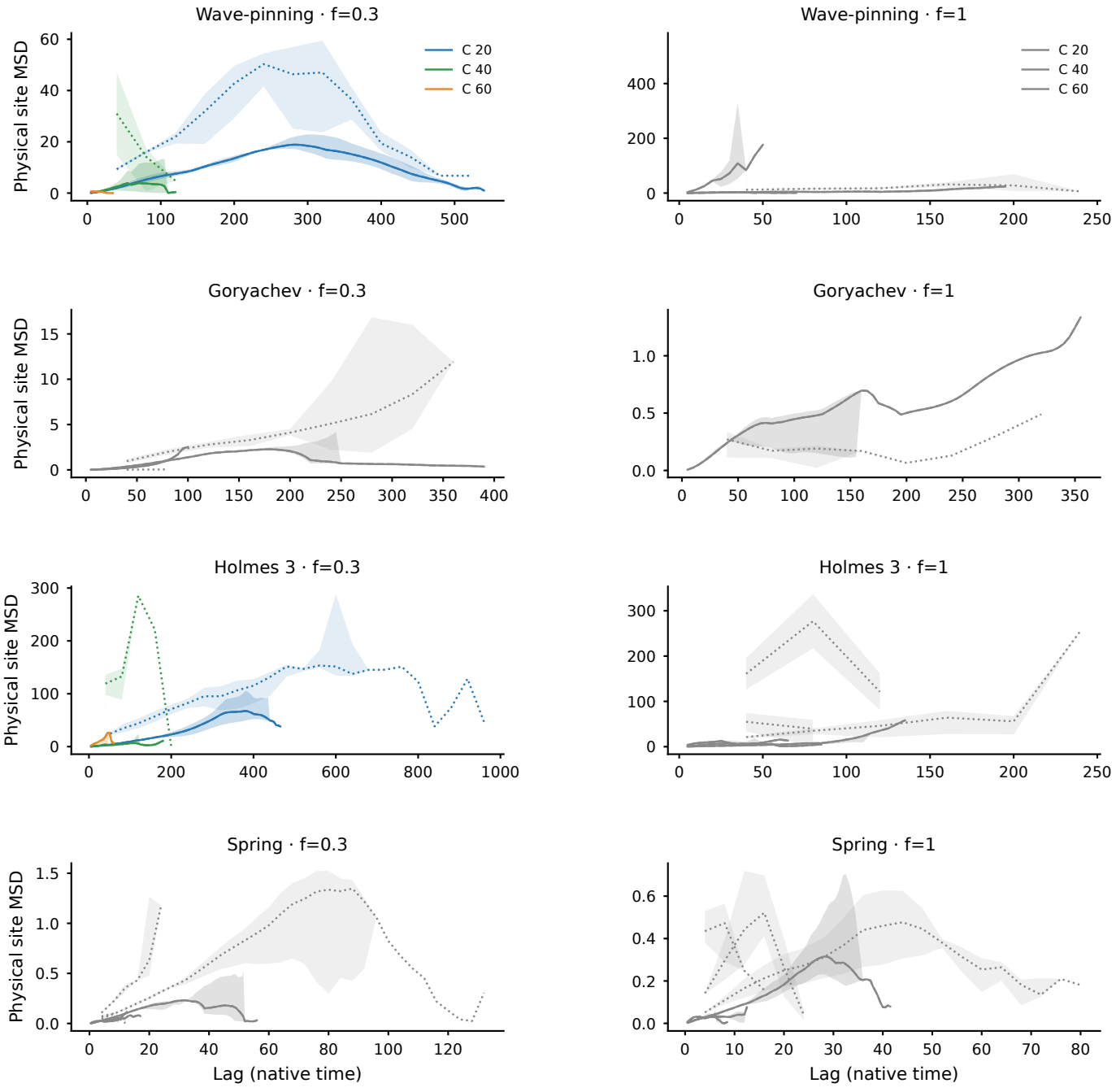
E0 INTERIM · A1–A4/A7 · provisional decisions · Solid: midpoint-C arc increments. Dashed: direct angle×actual-C sensitivity. Dotted: profile-peak sensitivity at coarse cadence. All methods reset at invalid frames; a plateau can indicate confinement.
CSV: figure4_E0_24_S5_2_siteMSD.csv

S5 Physical site motion in single-qualified-region tracks



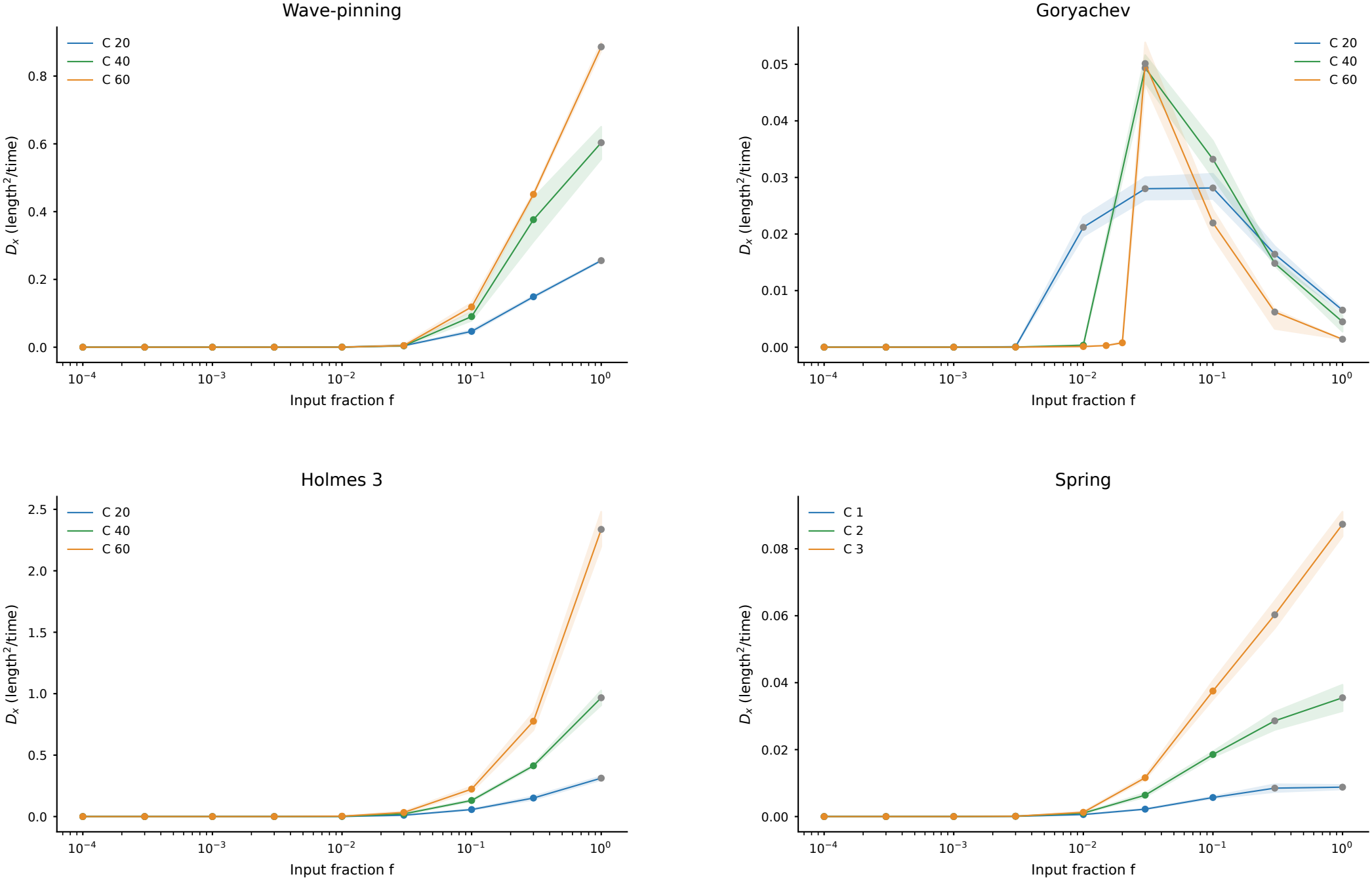
E0 INTERIM · A1–A4/A7 · provisional decisions · Solid: midpoint-C arc increments. Dashed: direct angle×actual-C sensitivity. Dotted: profile-peak sensitivity at coarse cadence. All methods reset at invalid frames; a plateau can indicate confinement.
CSV: figure4_E0_25_S5_3_siteMSD.csv

S5 Physical site motion in single-qualified-region tracks



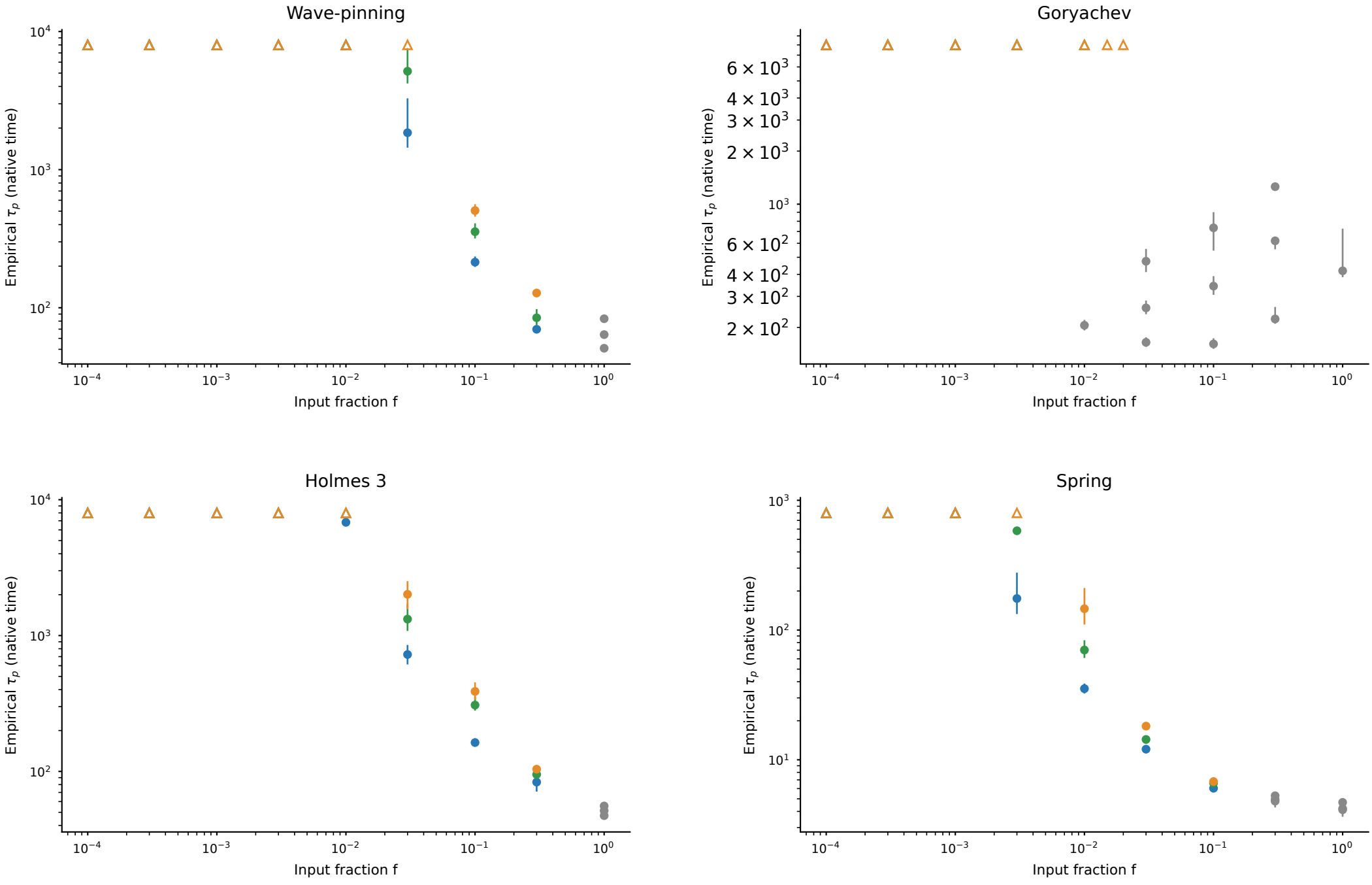
E0 INTERIM · A1–A4/A7 · provisional decisions · Solid: midpoint-C arc increments. Dashed: direct angle×actual-C sensitivity. Dotted: profile-peak sensitivity at coarse cadence. All methods reset at invalid frames; a plateau can indicate confinement.
CSV: figure4_E0_26_S5_4_siteMSD.csv

S5 Physical mobility across input strengths



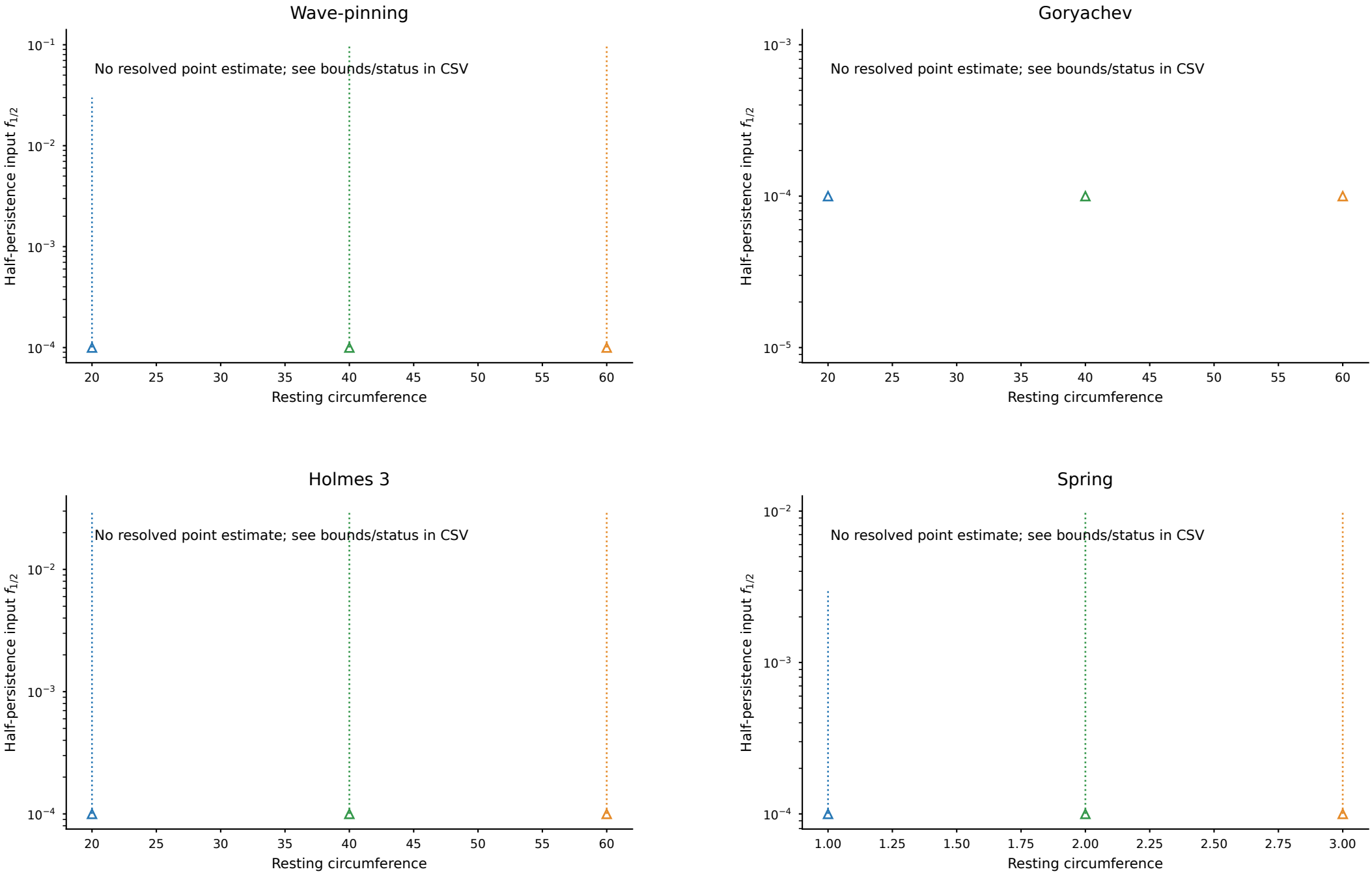
E0 INTERIM · A1–A4/A7 · provisional decisions · Dx = Dθ × (actual mean C / 2π)² is shown alongside the direct physical-track MSD, without assuming free diffusion.
CSV: figure4_E0_27_S5_Dx.csv

S6 Empirical persistence against input strength



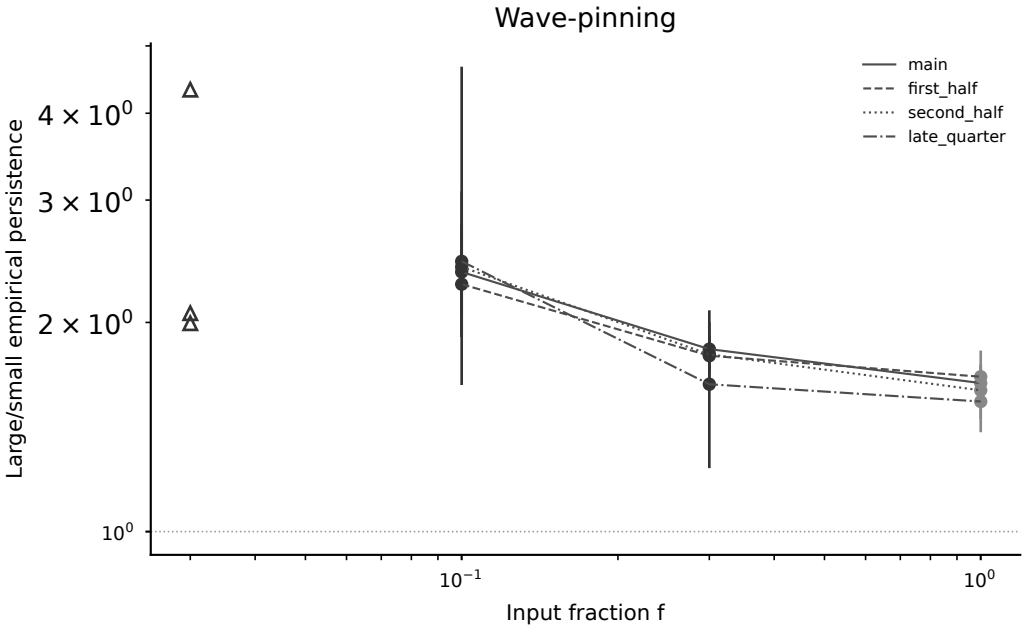
E0 INTERIM · A1–A4/A7 · provisional decisions · Upward triangles are lower bounds; downward triangles are unresolved below two cadence steps.
Exponential comparisons remain separate CSV columns and never replace empirical bounds.
CSV: figure4_E0_28_S6_tau.csv

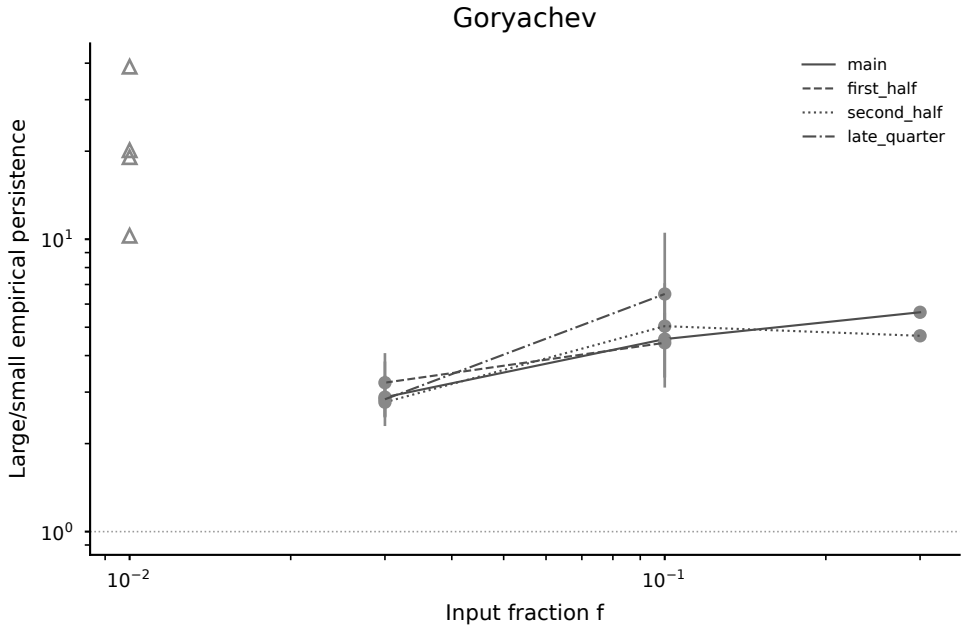
S6 Noise tolerance with finite-window uncertainty

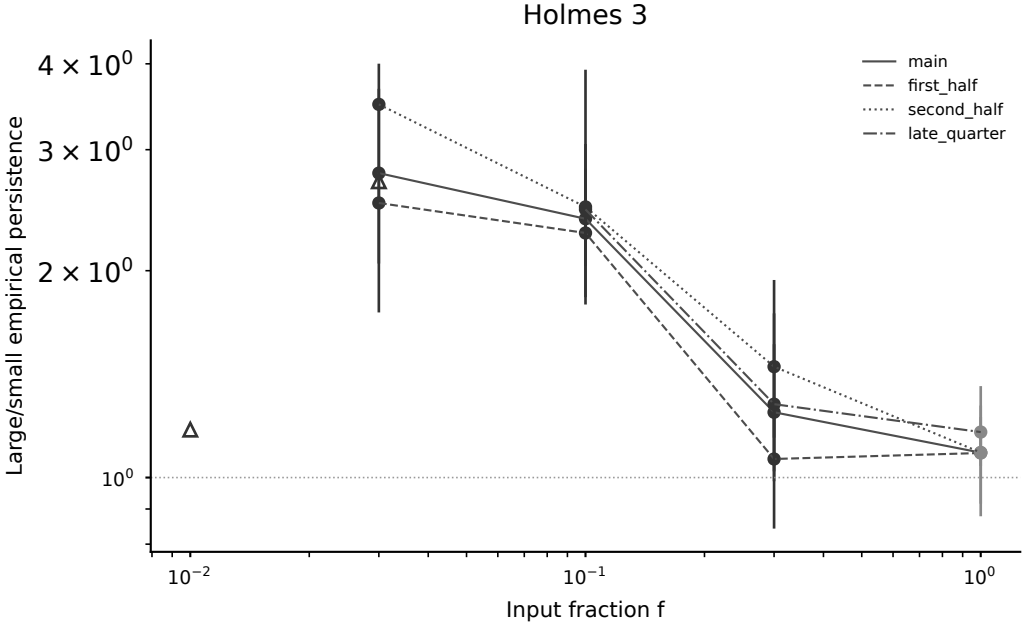


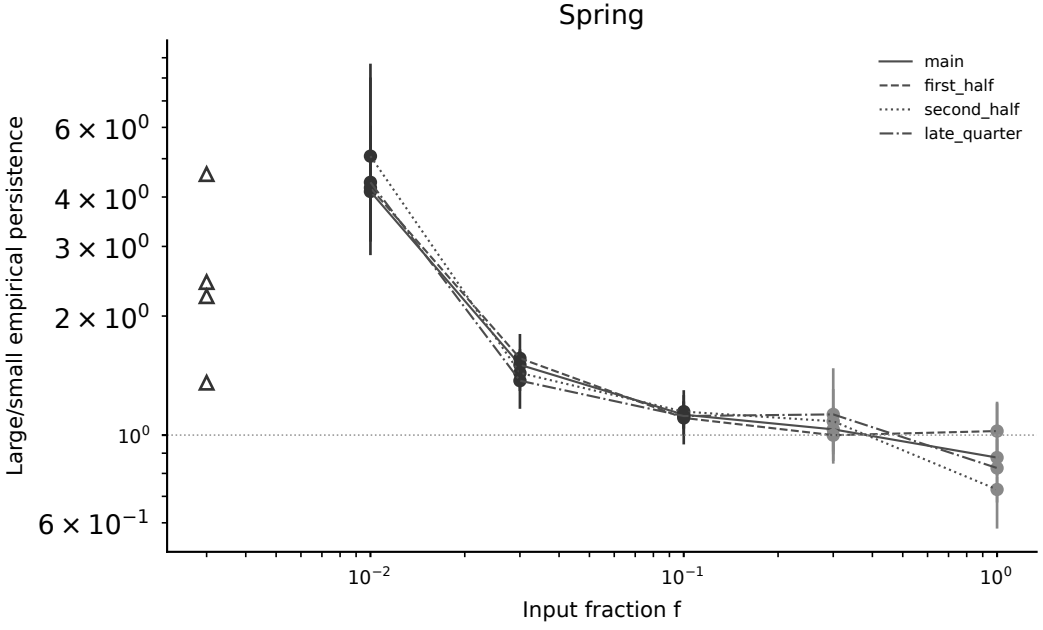
E0 INTERIM · A1–A4/A7 · provisional decisions · First crossing of half the weakest-input persistence, using log-f interpolation only when identified.
Censored weakest-input references remain bounded or inconclusive.
CSV: figure4_E0_29_S6_fhalf.csv

S12 Window robustness



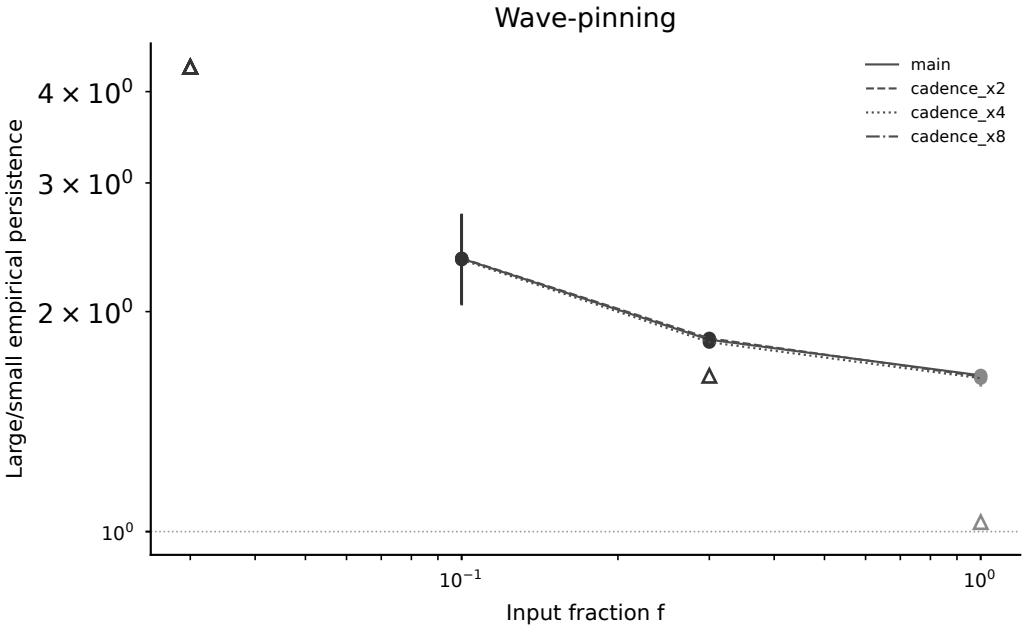


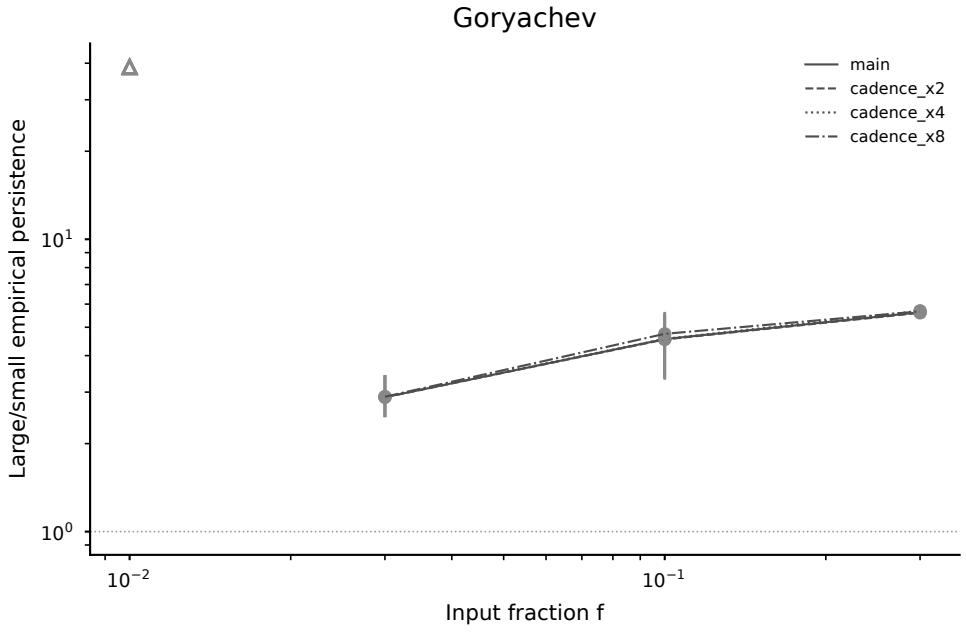


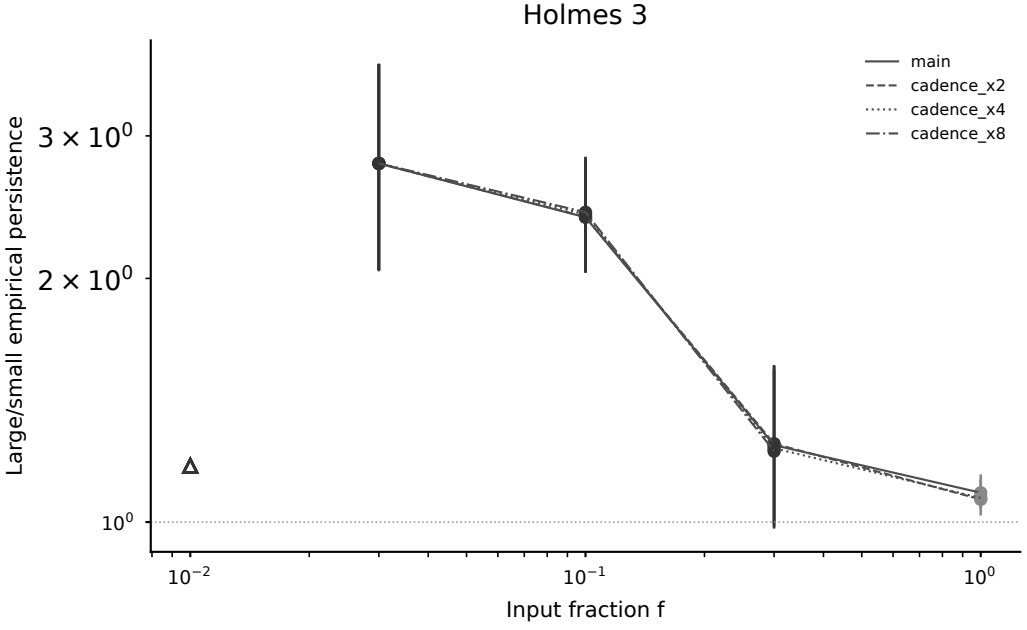


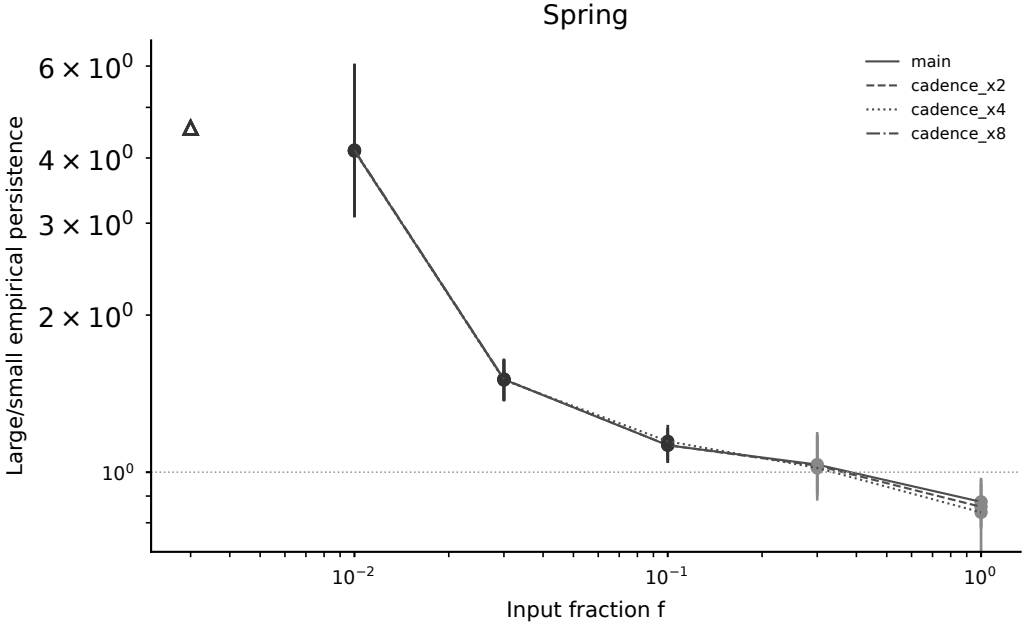
E0 INTERIM · A1–A4/A7 · provisional decisions · Each view is recomputed from its own valid pairs; the bootstrap unit remains the whole seed. Grey marks are retained forced conditions. E4 extended horizons are pending, so no extrapolated correction is shown.
CSV: figure4_E0_30_S12_1.csv

S12 Scalar-cadence robustness









E0 INTERIM · A1–A4/A7 · provisional decisions · Each view is recomputed from its own valid pairs; the bootstrap unit remains the whole seed. Grey marks are retained forced conditions. E4 extended horizons are pending, so no extrapolated correction is shown.
CSV: figure4_E0_31_S12_2.csv

S12 Non-forced decision sensitivity

Model	Window / cadence	H1	H2	Evaluable n	Reversal f	f½ CIs separate
Goryachev	main	inconclusive	inconclusive	4	none observed	no
Goryachev	first half	inconclusive	inconclusive	4	none observed	no
Goryachev	second half	inconclusive	inconclusive	4	none observed	no
Goryachev	late quarter	inconclusive	inconclusive	4	none observed	no
Goryachev	cadence x2	inconclusive	inconclusive	4	none observed	no
Goryachev	cadence x4	inconclusive	inconclusive	4	none observed	no
Goryachev	cadence x8	inconclusive	inconclusive	4	none observed	no
Holmes 3	main	inconclusive	provisional met	8	none observed	no
Holmes 3	first half	inconclusive	provisional met	8	none observed	no
Holmes 3	second half	inconclusive	inconclusive	8	none observed	no
Holmes 3	late quarter	inconclusive	inconclusive	8	none observed	no
Holmes 3	cadence x2	inconclusive	provisional met	8	none observed	no
Holmes 3	cadence x4	inconclusive	provisional met	8	none observed	no
Holmes 3	cadence x8	inconclusive	provisional met	8	none observed	no
Spring	main	inconclusive	inconclusive	7	none observed	no
Spring	first half	inconclusive	inconclusive	7	none observed	no
Spring	second half	inconclusive	inconclusive	7	none observed	no
Spring	late quarter	inconclusive	inconclusive	7	none observed	no
Spring	cadence x2	inconclusive	inconclusive	7	none observed	no
Spring	cadence x4	inconclusive	inconclusive	7	none observed	no
Spring	cadence x8	inconclusive	inconclusive	6	none observed	no
Wave-pinning	main	inconclusive	inconclusive	8	none observed	no
Wave-pinning	first half	inconclusive	inconclusive	8	none observed	no
Wave-pinning	second half	inconclusive	inconclusive	8	none observed	no
Wave-pinning	late quarter	inconclusive	inconclusive	8	none observed	no
Wave-pinning	cadence x2	inconclusive	inconclusive	8	none observed	no
Wave-pinning	cadence x4	inconclusive	inconclusive	8	none observed	no
Wave-pinning	cadence x8	inconclusive	inconclusive	7	none observed	no

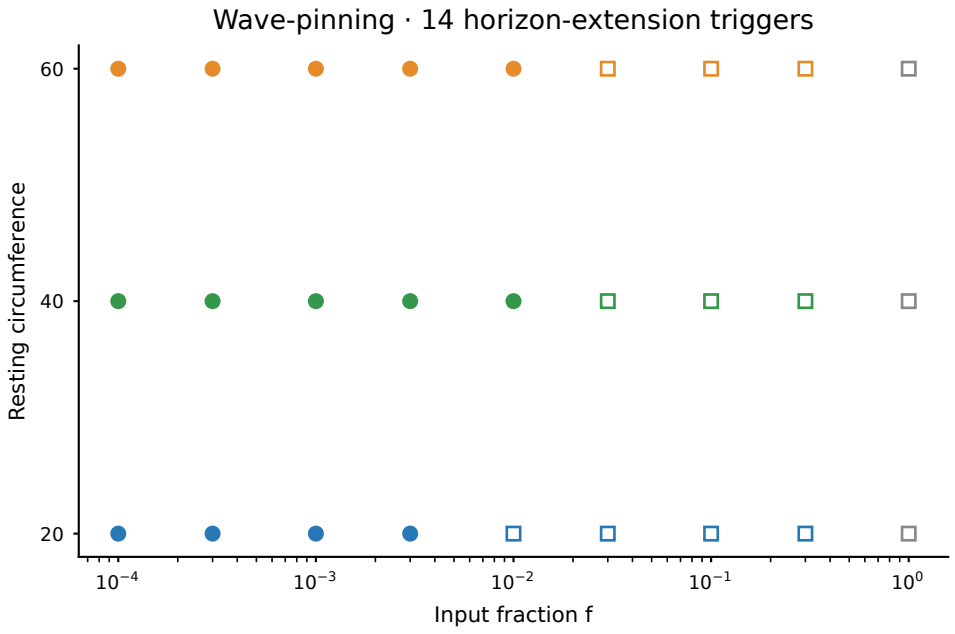
E0 INTERIM · A1–A4/A7 · provisional decisions · Provisional criteria are recomputed for every window/cadence. E1 forced-state continuation and the A6 overall H2 dependency are unavailable in E0.
CSV: figure4_E0_32_S12_decisions_nonforced.csv

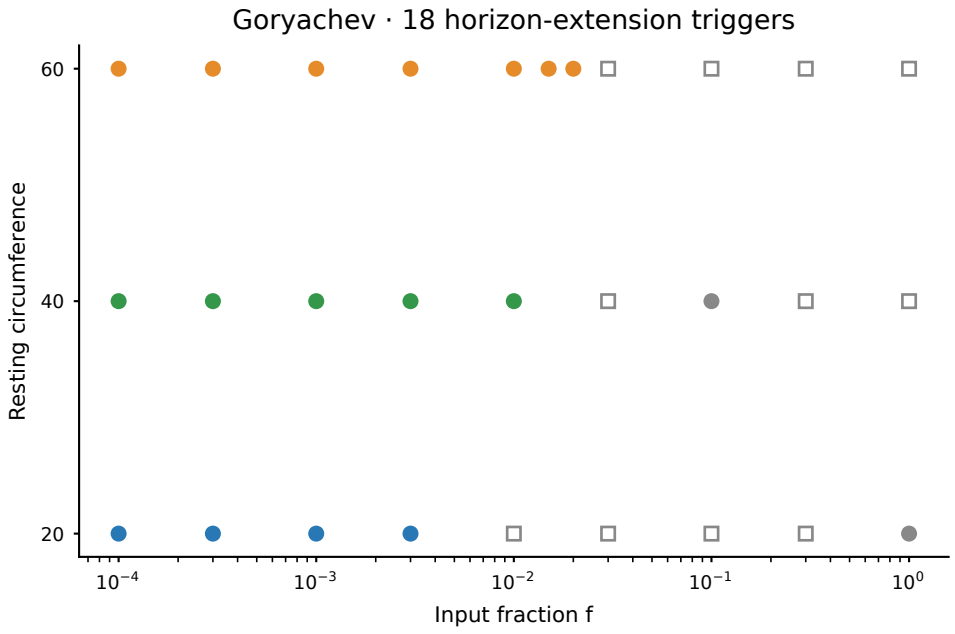
S12 Forced-inclusive decision sensitivity

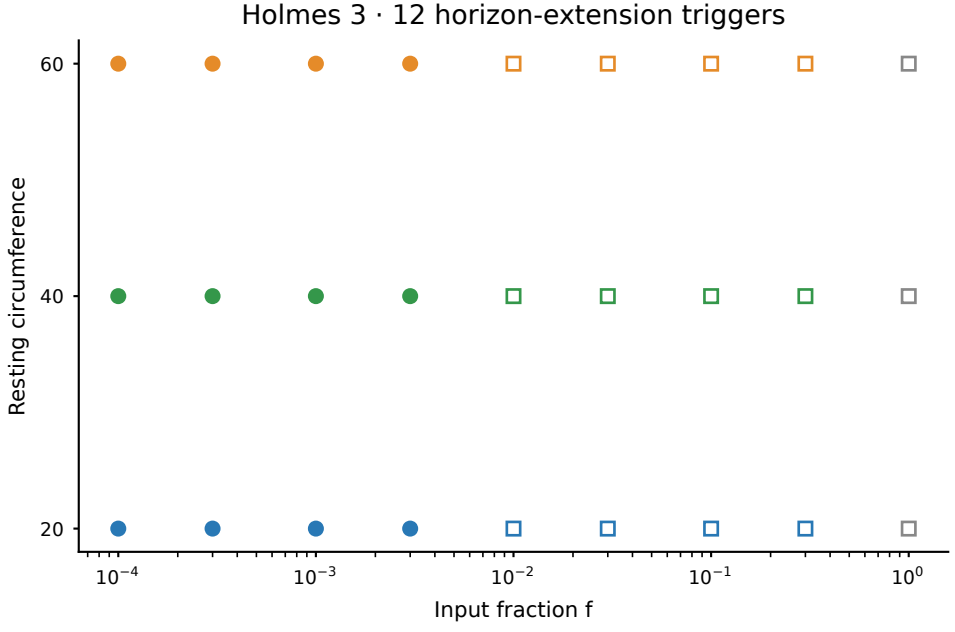
Model	Window / cadence	H1	H2	Evaluable n	Reversal f	f½ CIs separate
Goryachev	main	inconclusive	inconclusive	5	none observed	no
Goryachev	first half	inconclusive	inconclusive	5	none observed	no
Goryachev	second half	inconclusive	inconclusive	5	none observed	no
Goryachev	late quarter	inconclusive	inconclusive	5	none observed	no
Goryachev	cadence x2	inconclusive	inconclusive	5	none observed	no
Goryachev	cadence x4	inconclusive	inconclusive	5	none observed	no
Goryachev	cadence x8	inconclusive	inconclusive	5	none observed	no
Holmes 3	main	inconclusive	provisional met	9	none observed	no
Holmes 3	first half	inconclusive	provisional met	9	none observed	no
Holmes 3	second half	inconclusive	provisional met	9	none observed	no
Holmes 3	late quarter	inconclusive	inconclusive	9	none observed	no
Holmes 3	cadence x2	inconclusive	provisional met	9	none observed	no
Holmes 3	cadence x4	inconclusive	provisional met	9	none observed	no
Holmes 3	cadence x8	inconclusive	provisional met	8	none observed	no
Spring	main	criterion not met	provisional met	9	1	no
Spring	first half	inconclusive	inconclusive	9	none observed	no
Spring	second half	criterion not met	provisional met	9	1	no
Spring	late quarter	inconclusive	inconclusive	9	none observed	no
Spring	cadence x2	criterion not met	provisional met	9	1	no
Spring	cadence x4	criterion not met	provisional met	9	1	no
Spring	cadence x8	inconclusive	inconclusive	6	none observed	no
Wave-pinning	main	inconclusive	inconclusive	9	none observed	no
Wave-pinning	first half	inconclusive	inconclusive	9	none observed	no
Wave-pinning	second half	inconclusive	inconclusive	9	none observed	no
Wave-pinning	late quarter	inconclusive	inconclusive	9	none observed	no
Wave-pinning	cadence x2	inconclusive	inconclusive	9	none observed	no
Wave-pinning	cadence x4	inconclusive	inconclusive	9	none observed	no
Wave-pinning	cadence x8	inconclusive	inconclusive	7	none observed	no

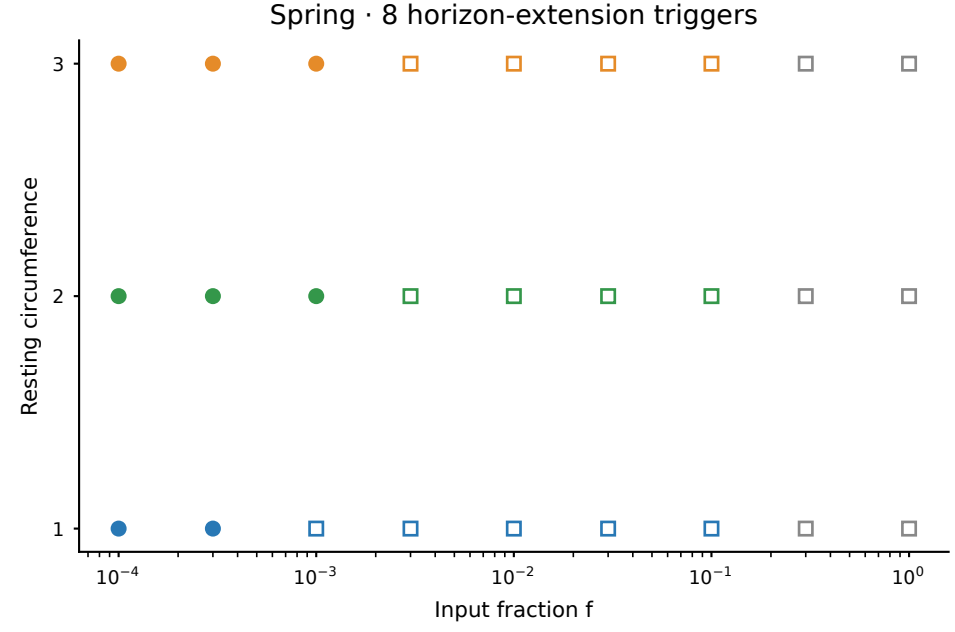
E0 INTERIM · A1–A4/A7 · provisional decisions · Provisional criteria are recomputed for every window/cadence. E1 forced-state continuation and the A6 overall H2 dependency are unavailable in E0.
CSV: figure4_E0_33_S12_decisions_inclusive.csv

S12 Finite-horizon limitations and pending extension inventory









E0 INTERIM · A1–A4/A7 · provisional decisions · Filled circles: cosine >0.8 at the longest usable lag; open squares: no trigger; crosses: no usable lag. All 110 groups remain shown, including forced conditions in grey. The 52 trigger groups require 624 proposed integrations at 12 seeds, and fourfold horizons do not guarantee uncensoring. These integrations have not entered E0.
CSV: figure4_E0_34_S12_E4_pending.csv